

ENTRY BARRIERS IN PROVIDER MARKETS: EVIDENCE FROM DIALYSIS CERTIFICATE-OF-NEED PROGRAMS

David A. Rosenkranz^a

September 14, 2024

([an updated version may be available here](#))

Abstract. How do entry barriers in provider markets affect market structure and welfare? In the US, certificate-of-need (CON) programs statutorily prohibit capacity investment when a health planner determines that a community’s “needs” are already being met. I examine CON programs’ effects in the dialysis industry, where patients are sensitive to treatment access and quality. I find that they reduce entry and protect incumbents from potential competition while enabling them to expand. I use variation from two natural experiments and a structural model of patient preferences to find that marginal entrants improve access and the patient-treatment match, lower congestion, and raise monthly countywide patient welfare by an amount equal to reducing travel by 9,254 miles.

Keywords: entry barriers, certificate-of-need, dialysis, access, congestion, health planning

JEL No.: I0, I1, I11, I18

^a Corresponding author. Department of Economics at Fordham University, 441 E Fordham Road, Dealy Hall, Bronx, NY 10458, USA. Email: drosenkranz@fordham.edu. Declarations of interest: none. I thank Douglas Almond, Sebastian Calonico, Ashley Swanson, and Ryan McDevitt. I also thank Guy Aridor, Tatyana Avilova, Michael Best, Norma Coe, Guy David, Daniel Deibler, Paul Eliason, Gautam Gowrisankaran, Teresa Harrison, Simon Lee, Stephen Lindner, Felipe Netto, Bozidar Plavsic, Silvio Ravaioli, Dario Romero, Marc Rysman, Pietro Tebaldi, Rachel Werner, and Bingxiao Wu. I am grateful to seminar attendees at numerous conferences, universities, and federal agencies, including the American Society of Health Economists’ Annual Conference, Bentley University, Columbia University, the Department of Justice, Fordham University, Hamilton College, the International Industrial Organization Society Conference, Johns Hopkins University, Washington University in St. Louis, Waseda University, and Wellesley College. All remaining errors are mine. The data reported here have been supplied by the United States Renal Data System (USRDS). The interpretation and reporting of these data are the responsibility of the author(s) and in no way should be seen as an official policy or interpretation of the U.S. government.

1 Introduction

Health care expenditures in the US are high and rising. Concerned that costs were being driven by moral hazard and provider-induced demand, lawmakers in the 1960s-70s established certificate-of-need (CON) programs to regulate health care capacity investment (Salkever 2000; Knuepfer 1974; Arrow 1963; Roemer 1961). Still widely used, they are now among America's oldest health care cost-containment initiatives. In states with CON programs, hospitals, nursing homes, and other providers seeking to open new facilities or expand existing capacity must receive permission from a state health planning agency. The agency may block any investments that it finds are "unnecessary."

In this paper, I investigate how CON programs affect market structure, treatment access, patient health, and welfare. Early studies compared states with and without CON programs and focused on hospitals. Their results vary, with some suggesting that CON programs are positively, negatively, and not significantly associated with hospital capacity and costs (e.g., Sloan and Steinwald 1980; Hellinger 1976; Salkever and Bice 1976). Recently, studies have begun using new sources of quasi-experimental variation to measure CON programs' effects on surgical quality, home health agencies, and MRI technology (e.g., Perry 2017; Polsky et al. 2014; Cutler et al. 2010).

I contribute to this literature causal estimates of CON programs' effects in the US dialysis industry. I study dialysis for several reasons. First, dialysis is a life-sustaining treatment for end-stage renal disease (ESRD) provided at specialized clinics called dialysis centers. Most individuals with ESRD are over 60 years old and suffer from other chronic conditions. For instance, they accounted for 1% of enrollees but 7.2% of expenditures in fee-for-service (FFS) Medicare in 2018 (USRDS 2020). Their poor health and dependence on nonresidential care may make their welfare and spending sensitive to CON-induced variation in access and quality, if any. Second, while critics of CON programs have argued that they dampen competition by "help[ing] incumbent firms amass or defend dominant market positions," there is little empirical evidence of this phenomenon to date (e.g., Ohlhausen 2015). The dialysis industry is highly concentrated, so if CON programs protect incumbents from potential entrants, then they may further weaken any remaining competitive pressure.

In particular, I contribute three main findings using novel sources of policy variation. First, I analyze the application process in North Carolina, where the NC dialysis CON program permits potential entrants to file applications for certificates of need when a simple function of public data

crosses a published threshold. I combine variation generated by these threshold-crossings with certificate of need application data to show that the NC dialysis CON program reduces entry, protects incumbents from potential competition, and enables incumbents to expand geographically.

Second, I measure how centers on CON programs' policy margins affect access, congestion, and health using data from the U.S. Renal Data System (USRDS). I combine the threshold-crossings in North Carolina with a 2007 reform in Washington that reduced its CON program's entry barriers. I use an instrumental variables difference-in-differences (IV-DID) research design. I find that on average, marginal centers reduce the distance between patients and their chosen centers by 8.670 miles (34% of the mean), and they reduce at-home dialysis utilization by 7.6 percentage points. Together, these findings indicate that CON programs reduce access enough to induce some patients to forgo travel (and the help of their center's staff). Marginal centers also lower congestion: on average, they increase the stations-per-patient and nurses-per-patient at patients' chosen centers by 0.105 (46%) and 0.038 (63%), respectively. Finally, I find evidence that marginal centers improve health on average, but the point estimates are not statistically significantly different from zero.

Third, I estimate a structural model of patient preferences to measure the marginal centers' effects on patient welfare. In the model, patients choose between nearby centers and at-home dialysis after considering their travel costs, switching costs, and alternative-specific fixed effects that capture all fixed unobserved quality characteristics. Because I do not observe patients make dollar-valued transactions, I measure their expected utility in miles-traveled equivalents (MTEs), which are the utility value of reducing distances to all alternatives by one mile. I combine the structural estimates with the IV-DID design to find that marginal centers increase expected utility by 359 MTEs per county-month on average. Since most patients receive dialysis thrice weekly, this is the value of reducing monthly countywide travel by 9,254 miles. Cost report data suggest that marginal centers' reported fixed costs are \$43,568 on average per month. A framework for evaluating the marginal centers' benefits and costs suggests that CON programs were blocking welfare-increasing entrants if the dollar value of an MTE is \$121 or more, though this threshold is likely an upper bound.

The remainder of this paper proceeds as follows. I describe dialysis, CON programs, and related literature in section 2. I describe the data in section 3. I compare states with and without dialysis CON programs in section 4. I analyze the NC dialysis CON program's application process in section 5. I conduct the IV-DID and welfare analysis in section 6. I conclude in section 7.

2 Background

2.1 Dialysis

Dialysis is the primary treatment for ESRD, a stage of chronic kidney disease characterized by permanent, life-threateningly low kidney function. Healthy kidneys remove waste from blood and contribute to red blood cell production. Without proper care, people with ESRD may become anemic and accumulate metabolic waste, causing them to feel sick within days and die within weeks (NKF 2024a, 2024b). There were up to 131,000 incident cases of ESRD per year in 2000-2019 and there were 565,000 prevalent cases in 2019 (USRDS 2022). ESRD is often caused by diabetes, high blood pressure, glomerulonephritis, and polycystic kidney disease (Mayo Clinic 2024b).

Dialysis treats ESRD by replacing normal healthy kidney function. There are two modalities. The first is hemodialysis (HD). During HD, a patient is intravenously connected to an external machine that filters their blood for 3-5 hours. The second modality is peritoneal dialysis (PD). During PD, dialysis fluid is poured through a catheter into a patient's peritoneal cavity, where it rests for a few hours and filters blood that passes by naturally. HD typically occurs in a dialysis center thrice weekly (in-center HD), but some patients receive home HD (HHD). PD typically occurs in a patient's residence or workplace daily. Regardless of modality, patients rely on a dialysis center for medication, training, monitoring, and supplies, though HHD and PD ("home dialysis," collectively) typically involve less travel and less assistance from medical professionals.¹ In 2019, 87% of people with ESRD used in-center HD, 11% used PD, and 2% used HHD (USRDS 2022).

A kidney transplant is the primary alternative to dialysis. People with ESRD who receive a new kidney have better prognoses and may no longer need dialysis (Mayo Clinic 2024b). For instance, the expected remaining lifetime of people ages 40 to 74 with ESRD in 2019 was 9-28 years with a transplant and 4-10 years otherwise (USRDS 2022). But donor kidneys are scarce: in 2022, there were only 25,550 transplants nationwide (MedPAC 2024). Consequently, most people with ESRD rely on dialysis (and dialysis centers) to survive.

¹There are some clinical and cost differences between HD and PD. For instance, PD patients may experience "longer lasting residual kidney function" but must undergo surgery to place the catheter (Mayo Clinic 2024a). Studies suggest that home dialysis is cheaper per-patient in the long term due to lower labor and overhead costs, but its initiation costs can make it costlier in the short term (Ferguson et al. 2021; Klomjit 2021; GAO 2015).

Dialysis centers are the primary care coordinators and treatment settings for people with ESRD. In 2022, there were 7,865 Medicare-certified centers nationwide. They were mostly freestanding (95%) and for-profit (89%), and had an average of 18 stations—units of capacity that produce one HD session at a time (MedPAC 2024). Their staff include social workers, dietitians, and at least one physician, but nurses and technicians provide most hands-on care. Monitoring patients (e.g., for hypotension) and sanitizing stations (e.g., to reduce infection rates) is labor intensive and subject to a quality-quantity trade-off (Grieco and McDevitt 2017). After years of consolidation, two firms—Davita and Fresenius—together owned 76% of centers in 2022 (MedPAC 2024; Wollmann 2024).

Medicare is a significant source of centers' revenues. In 2019, 84% of people with ESRD had FFS Medicare (65%) or Medicare Advantage (MA) plans (19%). Medicare's prominence in this market arises from the fact that all legal US residents with ESRD and a sufficient work history are Medicare-eligible, regardless of their age.² Private plans pay more on average but their rates vary considerably (e.g., League et al. 2022; Lin et al. 2022; Trish et al. 2021; Childers et al. 2019).

FFS Medicare pays dialysis centers a flat per-treatment rate that covers virtually all dialysis-related services, medications, equipment, and supplies (CMS 2019). The rate varies with patient characteristics (including age, body mass index, and comorbidities) and center characteristics (including location and facility-wide patient outcomes), includes add-ons for home dialysis training and high-cost outliers, but does not vary with treatment modality (Fed. Reg. 87 (214), 76 (3)). That the rate varies with facility-wide patient outcomes has lead some centers to provide low-scoring patients with additional care, or, in some cases, discharge them (Bertuzzi et al. 2023). Prior to 2011, prescription drugs were separately billable. Their addition to the payment bundle was associated with lower anemia drug (EPO) volume and fewer hospitalizations linked to EPO complications (Eliason et al. 2022; Swaminathan et al. 2012). Prior to 2005, the rate generally did not vary with patient characteristics (Leavitt 2008; MedPAC 2005, 2000). In 2019, it ranged from \$100 to \$1,000 per treatment and totaled \$10.7B—or \$29,000 per person with ESRD (League et al. 2022; USRDS 2022; Fed. Reg. 83 (220)).³ Beneficiaries pay a 20% coinsurance rate (CMS 2024b).⁴

²Coverage for people who are Medicare-eligible because of ESRD begins 3 months after ESRD onset (the “waiting period”). If they enroll, for the 30 months following the waiting period (the “coordination period”), Medicare is the primary payer if they have no private coverage; otherwise, Medicare is a secondary payer. Medicare is the primary payer after the coordination period (CMS 2023). Until 2021, people with ESRD generally could not enroll in MA, but could remain in their MA plan if they were already enrolled (Morgan and Kirchhoff 2021).

³Total FFS Medicare spending for people with ESRD for all services was approximately \$38.8B (USRDS 2022).

⁴This may be covered by a separate Medicaid or a Medigap plan, which I do not observe (MedPAC 2024).

2.2 Certificate-of-need programs

States began establishing CON programs in the 1960s, when health care costs were rising rapidly. National health expenditures grew from 5% of GDP in 1960 to 6.9% in 1970, an increase of nearly 40% (CMS 2024a). Medicare and Medicaid—which were established in 1965 as retrospective, cost-based payment systems—heightened concerns that providers were “paid [...] in a manner that provided virtually no efficiency incentives” (e.g., Salkever 2000). Policymakers were in “general agreement” that there was a surplus of health care resources whose costs were being borne indirectly by the public.⁵ For instance, in Illinois in 1974, State Senator Jack Knuepfer explained that:

[The] problem is essentially a surplus in hospital facilities. [...] There are presently plans [...] for almost 61 million dollars worth of new hospitals [...] and it is somewhat doubtful if [...] any [are] needed, since the present hospitals are not nearly full[.] [...] In my business, if I over expand [...] I go out of business. That is not what happens to a hospital. [...] [T]hose bills are paid by a third party payer and we care not one iota what those bills amount to. [...] So the problem of hospital economics is [...] the user cares not what it costs and does not and will not fight a cost increase. [Knuepfer 1974.]

Policymakers hoped that CON programs would cut costs by directly “controlling construction, controlling the amount of dollars that goes in” (Salkever and Bice 1979; Knuepfer 1974; Roemer 1961).⁶ New York established the first CON program in 1964 and other states soon followed. The National Health Planning and Resources Development Act of 1974 (NHPDA) created a federal mandate for CON programs, and every state but Louisiana had one by the 1980s. States with higher Medicaid spending established them sooner (e.g., Lanning et al. 1991). The NHPDA was repealed in 1987, but 35 states and Washington D.C. still had a CON program in 2024.⁷

In general, CON programs prohibit capital projects that do not meet a specific “community need.” However, they vary in how they determine whether a capital project is subject to review. For instance, depending on the state, they may review projects associated with hospitals, nursing homes, dialysis centers, or other health care institutions; projects associated with specific tech-

⁵Fed. Reg. 43(60): p. 13046 (March 28, 1978).

⁶Supporters of CON argue that their rents subsidize indigent care (e.g., Campbell and Fournier 1993; Posner 1971).

⁷For a fuller history and summary of CON programs, see e.g. NCSL (2024), AHPA (2016), Koopman and Philpot (2016), Ohlhausen (2015), AHPA (2006), and Blumstein and Sloan (1978). States continue to use Medicaid to justify their CON programs. For instance, Alaska explains that it “has a vested interest in new health care construction projects and equipment purchases because of the [...] money [it] expends for Medicaid” (ADOH 2024).

nologies, such as MRI scanners; or projects whose costs are forecast to exceed some minimum value. CON programs also vary in how they measure community need. For instance, some CON programs use publicly available formulas that combine published surveys of existing health care resources, population censuses, and estimates of per capita utilization rates; while others follow a less transparent process that culminates in a decision by a health planning commissioner.

Efforts by scholars to measure CON programs' effects have been complicated by potentially endogenous variation in their stringency, areas of focus, and designs, and the opacity of some of their decision-making processes.⁸ Early studies were largely descriptive and focused on hospitals. Their results varied considerably, with some suggesting that CON programs were positively, negatively, and not significantly associated with hospital capacity and costs (e.g., [Sloan and Steinwald 1980](#); [Hellinger 1976](#); [Salkever and Bice 1976](#)). Recently, studies have used variation generated by CON repeals, need determination formulas, and state boundaries to produce credible estimates of CON programs' effects on surgical quality, MRI scanners, and home health (e.g., [Perry 2017](#); [Polsky et al. 2014](#); [Cutler et al. 2010](#)). States with dialysis CON programs have been associated with less dialysis capacity in the 1980s and 2007, but little is known about their casual effects, including on access, health, and welfare ([Dai and Tang 2015](#); [Dai 2014](#); [Ford and Kaserman 1993](#)). I contribute estimates of these effects generated by within-place variation in CON programs' strenuousness.

In theory, CON programs may also dampen competition by “help[ing] incumbent firms amass or defend dominant market positions” (e.g., [Ohlhausen 2015](#)). Policymakers have said that hospitals had “ask[ed] to be put under this regulation,” and CON programs were established sooner in states where market concentration may have helped form lobbying coalitions ([Wendling and Werner 1980](#); [Knuepfer 1974](#)).⁹ Recent studies show that states that review imaging technology purchases have wider disparities between older and newer hospitals' use of MRI, CT, and PET scans in favor of older hospitals, and home health quality is less sensitive to public reporting in CON states ([Baker and Stratmann 2021](#); [Wu et al. 2019](#)). I contribute empirical evidence of the statutory mechanism that protects incumbents from potential competition while enabling them to expand.

⁸For instance, [Russell \(1979\)](#) suggested that “sometimes it is possible to get around the review [...] by [...] splitting the project[.]” [Eastaugh \(1982\)](#) “attempt[ed] to improve the [...] precision of the [...] CON [variable],” but found “that variability in programme implementation was high[;]” and [Campbell and Fournier \(1993\)](#) observed that “[r]egulators [...] enjoy much discretion[.]” Other concurrent policy changes further complicated early studies (e.g., [Sloan 1981](#)).

⁹Market participants claim that CON programs facilitate anticompetitive conduct, and some have allegedly successfully appealed to CON programs to withhold certificates-of-need from their competitors ([DOJ & FTC 2008, 2004](#)).

3 Data

3.1 Data sources

The primary data for this study are from the U.S. Renal Data System 2018 Database (USRDS Database) ([USRDS 2018](#)). The USRDS Database contains standard analysis files (SAFs) derived from CMS data, including Medicare claims, the chronic renal disease medical evidence reports, death notification reports, and facility surveys. It also contains a 100% sample of institutional Medicare claims for Medicare beneficiaries with renal disease. Other vintages of the USRDS Database have been used elsewhere (e.g., [Wollmann 2024](#); [Eliason et al. 2022](#); [Eliason et al. 2020](#)).

I use the USRDS database and some ancillary public datasets to create a sample of patient-months spanning 1980-2016. A patient-month is included in the sample if the patient was receiving dialysis in that month. For each patient-month, I observe a rich set of data encompassing their dialysis treatments, demographics, clinical conditions, residences, insurance status, hospitalizations, Medicare spending, and travel to-and-from dialysis centers. I also use the USRDS Database and ancillary data to create a sample of facility-months spanning 1980-2016. A facility-month is included in the sample if it treated at least one patient in that month. For each facility-month, I observe the center's volume, business model, chain affiliation, location, and staffing.¹⁰

I also use data from the NC dialysis CON program. First, I use data published in semiannual dialysis reports (SDRs) in 1997-2019 that the NC dialysis CON program used to make need determinations. Second, I use contemporaneous records of applications for certificates-of-need. For each application and its associated project, I observe key dates, the applicant's name, the result of the review process, and the project's location, projected cost, and purpose. I use these data to analyze the NC dialysis CON program's need determination and application process.

Finally, I use data from CMS's Healthcare Cost Report Information System (HCRIS) spanning 2011-2016. HCRIS contains records of Medicare-certified institutional providers' annual cost reports. For each center, I observe several annual measures of reported labor, capital, and administrative costs. I use these data to incorporate centers' reported fixed costs in the welfare analysis.

See appendix [A](#) for more information about the key variables and data sources.

¹⁰Volume varies monthly but other facility-level variables vary yearly. Staffing is available starting in 2004.

3.2 Descriptive statistics

In this section, I summarize the data for years spanning 2004-2016 to align with the cross-sectional analysis in section 4. Appendix B reports descriptive statistics for the sample spanning 1980-2016.

Table 1 describes the distribution of patients and centers over counties. In column (1), it shows that there was at least one center in 55% of county-months. It also shows that on average in a given month, there were 2.10 centers, 34.79 stations, and 127 resident patients per county.

Table 2 describes patients' pre-dialysis characteristics. In column (1), it shows that patients were 61 years old on average, 65% White, 29% Black, 14% Hispanic, and 56% male. At the onset of ESRD, 11% of patients had some employment and, on average, they had a BMI of 29. The primary cause of ESRD was diabetes for 44% of patients and hypertension for 28% of patients.

Table 3 describes patients' treatment choices and choice sets. In column (1), it shows that on average in a given month, patients had 161 centers within 100 miles of their residence. Thirty percent (49.25) were owned by independent providers, and the rest were split between 5.23 chains. The nearest center was located 3.73 miles away and the nearest center with above-median stations-per-patient was located 7.80 miles away. Patients' chosen centers were 10.47 miles away, had 0.25 stations-per-patient, and 0.07 nurses-per-patient. Only 2% of patients switched centers in a given month on average. Only 14% of patients ever received a transplant.

Table 4 describes patients' health outcomes, payers, and Medicare spending. In column (1), it shows that on average in a given month, 70% of patients had Medicare FFS as a primary payer. Among these patients, I observe the urea reduction ratio (URR). Typically, URR is increasing in session duration and should remain above 65% (NKUDIC 2009). On average in a given month, 8% of patients with Medicare FFS as a primary payer had a URR below 65%. Seventeen percent had a hospital stay and their average monthly Medicare spending was \$5,313.

Finally, table 5 describes centers' characteristics. In column (1), it shows that on average in a given year, 79% were for-profit and 57% were owned by Davita or Fresenius. On average in a given month, they served 69.80 patients, operated 0.77 stations-per-patient, and were located 8.29 miles from a competing center. Since 2011, their total monthly reported costs were \$224,793 on average—split between capital (\$36,960), staffing (\$69,514), supplies (\$23,928), administration (\$54,695), and other (\$39,695). Their monthly reported costs per patient were \$3,375, on average.

4 Cross-sectional variation between CON and non-CON states

In this section, I compare states with and without dialysis CON programs (CON and non-CON states). In the spirit of Horwitz et al. (2024), Wu et al. (2019), Polsky et al. (2014), and others, I also compare counties along CON state borders. I focus on 2004-2016, when surveys indicate that AL, AK, HI, IL, ME, MS, NY, NC, VT, WA, DC, and WV were CON states.¹¹

4.1 Estimation strategy

Define indexes i for units, t for periods, and c for counties. Let $c(i, t)$ be unit i 's county in period t and let $\text{CON}(c)$ indicate that county c is in a CON state. I estimate linear regressions of the form:

$$Y_{it} = \beta \text{CON}(c(i, t)) + \text{FE}_t + \varepsilon_{it} \quad (1)$$

where Y_{it} is an outcome, FE_t is a period fixed effect, and β measures statewide regression-adjusted average outcome differences between CON and non-CON states.

In the border analysis, further define the index e for counties in CON states adjacent to non-CON states. For each county e , let $C(e)$ be a set consisting of e and its adjacent counties in non-CON states.¹² The index e now also identifies groups of counties on state borders. I construct a sample of (e, i, t) -level observations such that $c(i, t) \in C(e)$ and estimate linear regressions of the form:

$$Y_{eit} = \beta \text{CON}(c(i, t)) + \text{FE}_{et} + \varepsilon_{eit} \quad (2)$$

where Y_{eit} is an outcome, FE_{et} is a group-specific period fixed effect, and β measures regression-adjusted average outcome differences between adjacent counties on CON state borders.¹³

I use state-level cluster-robust standard errors because $\text{CON}(\cdot)$ varies by state. In the border analysis, this accounts for duplicate data arising from counties in multiple border county groups.

¹¹Surveys conducted by the American Health Planning Association in 2004, 2005, 2010, 2012, and 2016 indicate that these 11 states had dialysis CON programs in each survey year and DC had a dialysis CON program in each survey year except 2005. However, I have found no evidence that DC ended and reinstated its dialysis CON program between 2004 and 2010. Therefore, I disregard DC's 2005 omission. I assume that each state's status as a CON or non-CON state was unchanged in the intervening years. I have not found comparable surveys prior to 2004.

¹²For instance, if e is White, IL, then $C(e)$ consists of White, IL, Gibson, IN and Posey, IN.

¹³Figure A1 plots CON states and their border counties.

4.2 Results

In column (4), table 1 shows that counties in CON states had 0.06 more centers and 2.77 more stations on average in a given month. While these estimates are not statistically significant, they apparently show that counties in CON states may have had *more* dialysis-related capital resources. This surprising finding is explained by the states' patient populations: counties in CON states had 30.25 (25%) more resident patients on average in a given month. Though this is also not statistically significant, it suggests equations (1) and (2) should include county characteristics. I interact the month fixed effects with the size of each county's resident dialysis patient population hereafter. In column (5), table 1 shows that counties in CON states had 0.33 fewer centers and 4.52 fewer stations than counties in non-CON states with a similar number of resident patients. In columns (6)-(9), it reports qualitatively similar differences at state borders.

Tables 2-5 present results for patient and center characteristics. While some results are consistent with theory (e.g., that patients in CON states are treated at centers with more congestion, or that centers in CON states are older on average), they are often not statistically significant.

The cross-sectional analysis has two core limitations in this context. First, statistical inference is complicated by a lack of sharp variation in $CON(\cdot)$. This limitation arises from the fact that only eleven states and DC had dialysis CON programs in 2004-2016, that these places were concentrated in the Eastern United States, and that this analysis abstracts from policy features that may vary the strenuousness and scope of dialysis CON programs within and between states.

Second, causal inference is complicated by potential confounding from other, unrelated differences between states, including potential differences in construction costs or baseline patient health. While the border analysis isolates variation between adjacent counties, some confounding may remain. For instance, table 1 shows that border counties in CON states have 46.78 (28%) more resident patients, on average. Furthermore, other government programs also vary at the state level.¹⁴ That the cross-sectional analysis abstracts from specific policy features complicates efforts to directly tie differences between states to CON programs per se or investigate mechanisms.

In the remainder of the paper, I overcome these limitations by analyzing within-place quasi-experimental variation in the WA and NC dialysis CON programs' strenuousness.

¹⁴For instance, Medicaid varies by state and table 4 shows that 36% of patients were dual-eligible in an average month. Moreover, all states with a dialysis CON program also had other CON programs in 2016 (AHPA 2016).

5 Evidence from the application process in North Carolina

While the cross-sectional analysis showed that counties in CON states had fewer dialysis-related capital resources than other counties with similar resident patient populations, it is not clear whether this difference is causally attributable to CON. And while critics of CON argue that it dampens competition by “help[ing] incumbent firms,” there is comparatively little empirical evidence of this phenomenon (e.g., [Ohlhausen 2015](#)). In this section, I combine application data with variation generated by minutiae in how the NC dialysis CON program determines when a new center is needed. I establish a causal link between the NC dialysis CON program and market structure, and I describe how incumbents can use it to all but eliminate competition from potential entrants.¹⁵

The NC dialysis CON program determines when a county needs a new center using two threshold rules.¹⁶ First, it computes how many additional stations are needed to serve a 6-month forecast of each county’s resident in-center dialysis patient population assuming that each station serves 3.2 patients (“station deficit”). To illustrate, if a county with 9 stations was forecast to have 64 patients, then its station deficit would be $\frac{64}{3.2} - 9 = 11$.¹⁷ Second, the NC dialysis CON program measures each center’s patient-to-station ratio and identifies the lowest ratio in each county, if any (“minimum utilization rate”). A new center may open in a county if the county’s station deficit is at least 9.5. If the county has existing centers, then its minimum utilization rate must also be at least 3.2. Applications to open new centers can be filed semiannually after each SDR’s publication.

In counties with existing centers (“incumbents”), these rules create two mechanisms for incumbents to block potential entrants. First, incumbents can ensure that their county’s station deficit is always less than 9.5 by investing in new stations as their county’s resident in-center dialysis patient population grows. This is feasible under the NC dialysis CON program’s rules for capacity investment.¹⁸ Second, incumbents may ensure that their county’s minimum utilization rate is always less than 3.2 by treating fewer than 3.2 patients-per-station at one of their centers.

There is a clear financial incentive for incumbents to exercise these opportunities to block po-

¹⁵I focus on North Carolina because it is uncommonly transparent about the CON process and its rules feature thresholds that enable a clear presentation of the statutory mechanism and its implications.

¹⁶The need determinations are made in “planning areas,” which are generally counties. I refer to them as counties.

¹⁷The forecast is based on each county’s patient population and lagged 5-year growth rate.

¹⁸Existing centers can add new stations if they are treating at least 3.2 patients per station and they are projected to need at least one additional station to serve a 6-month forecast of their own in-center dialysis patient population.

tential entrants. First, dialysis centers compete on quality—e.g., cleanliness, staff attention, and comfort—in markets where patients can select from several nearby alternatives (e.g., [Eliason 2022](#); [Grieco and McDevitt 2017](#)). They may also compete on price among commercial insurance plans. Second, the NC dialysis CON program permits incumbents to spin off existing stations to new locations in the same county—even if doing so would result in the establishment of a new center. Consequently, incumbents can cut quality investment costs, raise prices, and expand geographically by leveraging the NC dialysis CON program to block potential entrants.

But do incumbents exercise these opportunities? Are the thresholds binding? To answer these questions, I use data in the SDRs to compute each North Carolina county’s station deficit and minimum utilization rate for all half-years in 1997–2019. I also use the NC dialysis CON program’s application data to identify when someone filed an application to open a new center. Filing applications for certificates-of-need is costly. They are sometimes hundreds of pages long, include patient testimonials, data analysis, construction plans, and financial documents, and require an application fee. Filing an application is therefore a strong signal of an applicant’s intention to open a new center.

In counties without incumbents in 1997, figure 1(a) plots the rate at which potential entrants filed applications to open centers as a function of their counties’ station deficits. It shows that filing rates move from nearly 0% to nearly 100% when county station deficits exceed 9.5.¹⁹ Figure 1(b) plots the counties’ station deficits over time. It shows that threshold-crossings consistently predict application-filing dates. Recall that in a county without incumbents, the station deficit exceeds 9.5 when a 6-month forecast of the county’s resident in-center dialysis patient population exceeds $9.5 \times 3.2 = 30.4$. There is no clear reason why this threshold would be discontinuously related to filings unless applicants understood that the NC dialysis CON program’s rules were binding.

In county-half-years with incumbents, figure 2 plots the joint distribution between station deficits and minimum utilization rates. Panel (a) shows that potential entrants file applications in 82% of county-half-years when station deficits and minimum utilization rates exceed their respective thresholds—and, often, just barely. But it also shows that these conditions are met only 0.6% of the time because incumbents generally keep station deficits low. Meanwhile, panel (b) shows that incumbents commonly expand geographically by applying to spin off stations to new locations, including when potential entrants are statutorily prohibited from entering.

¹⁹In appendix C, I use an RD to find that threshold-crossings cause filing rates to increase by 55 percentage points.

6 The causal effect of centers on CON programs' margins

6.1 Natural experiments in North Carolina and Washington

The foregoing section showed that the NC dialysis CON program's threshold rules are binding on applications to open new centers. In this section, I leverage natural experiments in North Carolina and Washington to confirm that entry follows after the NC and WA dialysis CON programs relax entry limits. I also estimate how entrants on their policy margins affect patient welfare.

In North Carolina, I leverage variation generated in counties without incumbents when I observe their station deficits first exceed 9.5. The timing and circumstances of these threshold-crossings are plausibly exogenous because they depend only on the counties' resident in-center dialysis patient population sizes and growth rates. While counties with incumbents also sometimes experience threshold-crossings, the timing and circumstances of their threshold-crossings are endogenous to the incumbents' behavior and may therefore coincide with other, unobserved market-level shocks.

In Washington, the WA dialysis CON program uses a similar process to determine whether new centers are needed. Like the NC dialysis CON program, it estimates how many stations are needed to serve a forecast of each county's resident in-center dialysis patient population, and it considers whether any incumbents' utilization rates are low. One important difference is that there is no station deficit threshold: a new center can open to meet a county's need for additional stations even if that need is small. Therefore, while centers' incentives are likely to be similar in Washington and North Carolina, I am unable to use variation generated by threshold-crossings in Washington.

Instead, I use variation generated by a 2007 reform. The reform made entry statutorily easier in two ways in counties that did not have incumbents in 2006 (the "target WA counties"). First, before 2007, the WA dialysis CON program estimated station deficits assuming that each station could serve 4.8 patients. The reform reduced this parameter to 3.2 in the target WA counties. Second, the reform enabled the WA dialysis CON program to estimate a target WA county's station deficit using its own population and the populations of adjacent target WA counties. Figure [A2](#) shows that many target WA counties neighbored one another. Collectively, these changes increased station deficits in the target WA counties, thereby enabling potential entrants to open there with more stations, sooner.

I discuss further details of the 2007 WA reform and the NC threshold-crossings in appendix [D](#).

6.2 Identifying the effect of a marginal dialysis center

I leverage variation generated by the foregoing natural experiments using an instrumental variables difference-in-differences research design (IV-DID). I regard the 2007 WA policy change and the NC threshold-crossings as events that suddenly reduce regulatory entry costs in the event counties. The comparison counties with respect to each event are other counties in the US without a dialysis center prior to that event. I make the following assumptions to draw causal inferences in this context:

A1: Exclusion restriction. The events did not affect outcomes except through their effect on entry. This assumption is plausibly satisfied because the events were administrative decisions that statutorily relaxed dialysis centers' entry barriers and had no other obvious economic or health impacts.

A2: Relevance and monotonicity. The events sometimes made entry more likely, and never made it less likely. This assumption is plausibly satisfied because the events statutorily made entry easier for centers with more stations (in WA) or at all (in NC).

A3: First stage mean independence. Post-event entry rates in the event and comparison counties would have been equal but for the events. This assumption would be satisfied if the average net benefits of entry in the event and comparison counties would have moved in parallel but for the events (e.g., if CON programs raised the level of entry costs, but left time trends unchanged).

A4: Reduced form no anticipation and parallel trends. The events did not affect average pre-event outcomes in the event counties, and average outcomes in the event and comparison counties would have moved in parallel but for the events. This assumption is plausibly satisfied in NC, where the events occurred when forecasted patient populations happened to exceed $9.5 \times 3.2 = 30.4$. In WA, it would be violated if the reform was spurred by declining health or welfare in the event counties. I will therefore examine pre-event outcome trends in the event and comparison counties.

Under these assumptions, IV-DID identifies the average treatment effect of a dialysis center that opened in an event county because of the event. This is the complier average treatment effect at the event site (CATE-ES).²⁰ The first stage compares entry trends in the event and comparison counties before-and-after the events. The reduced form does likewise for outcome trends. Their ratio identifies the CATE-ES (e.g., [Duflo 2001](#)). See appendix E for details.

²⁰This term combines IV's "complier average treatment effect" with DID's "average treatment effect on the treated." I use "at the event site" instead of "on the treated" because in IV-DID, the event is the instrument, not the treatment.

6.3 Estimating the effect of a marginal dialysis center

I estimate the CATE-ES using two-stage least squares (2SLS). I address concerns about DID estimation with two-way fixed effects using the stacked regression (Gardner et al. 2024, Goodman-Bacon 2021, Cengiz et al. 2019). Define the indexes e for events, i for patients, c for counties, and t for months. Let $c(i, t)$ and $z(i, t)$ identify patient i 's county and ZIP code in month t , respectively. I duplicate the patient-month sample described in section 3 once for each event to generate an (event, patient, month)-level dataset. I keep only those observations (e, i, t) such that $c(i, t)$ is an event or comparison county for e and t is within 72 months of e 's effective date.²¹ That is, for each event, this dataset is a 144-month patient-level repeated cross-section of the event and comparison counties.

Let $D_{eit} := 1[c(i, t) \text{ has a center}]$. Let τ_e be e 's effective date and let the instrument for D_{eit} be $Z_{eit} := 1[c(i, t) \text{ is } e\text{'s event county}]1[t \geq \tau_e]$. I estimate 2SLS models of the form:

$$Y_{eit} = \beta D_{eit} + \Gamma_e X_{eit} + \text{FE}_{ez(i,t)} + \text{FE}_{et} \times \text{PatPop}_{ec(i,t)t} + \varepsilon_{eit} \quad (3a)$$

$$D_{eit} = \delta Z_{eit} + \Omega_e X_{eit} + \text{FE}_{ez(i,t)} + \text{FE}_{et} \times \text{PatPop}_{ec(i,t)t} + \eta_{eit} \quad (3b)$$

where Y_{eit} is an outcome, β is the parameter of interest, X_{eit} is a vector of control variables, $\text{FE}_{ez(i,t)}$ is an event-specific ZIP code fixed effect, and $\text{FE}_{et} \times \text{PatPop}_{ec(i,t)t}$ is an event-specific month fixed effect fully interacted with the county's resident patient population.

Finally, let $\mathcal{W} := \{-6, \dots, 5\}$ contain annual leads and lags excluding -1. For every $\tau \in \mathcal{W}$, let $Z_{eit\tau} := 1[c(i, t) \text{ is } e\text{'s event county}]1[(t - \tau_e)/12] = \tau]$. I also estimate models of the form:

$$Y_{eit} = \sum_{\tau \in \mathcal{W}} \alpha_\tau Z_{eit\tau} + \Lambda_e X_{eit} + \text{FE}_{ez(i,t)} + \text{FE}_{et} \times \text{PatPop}_{ec(i,t)t} + \nu_{eit} \quad (4)$$

$$D_{eit} = \sum_{\tau \in \mathcal{W}} \delta_\tau Z_{eit\tau} + \Xi_e X_{eit} + \text{FE}_{ez(i,t)} + \text{FE}_{et} \times \text{PatPop}_{ec(i,t)t} + \omega_{eit} \quad (5)$$

where $(\alpha_\tau : \tau \in \mathcal{W})$ and $(\delta_\tau : \tau \in \mathcal{W})$ measure the dynamic effects of Z on Y and D , respectively.

I use county-level clustered standard errors because the events vary by county. This accounts for duplicate data arising from stacking. I also report 95% VtF critical values (Lee et al. 2023).²²

²¹I assume that each event's effective date was 18 months after the event because it takes time to open a new center. I chose 18 months because that is the earliest that a new center opened in the event counties after one of the events.

²²The VtF critical values enable conducting weak instrument-robust inference without using an F-statistic threshold.

6.4 IV-DID results

6.4.1 Dialysis CON programs' effects on entry

Figure 3 plots the estimated dynamic first stage effect of the events on entry. It shows that on average in a given month after the events, patients in event counties were 53% more likely to have a center in their county. In North Carolina, this is consistent with my earlier finding that threshold-crossings caused potential entrants to file applications to open new centers. In Washington, I investigate the link between the reform and entry by checking whether the centers that opened in the target WA counties could have opened under the pre-2007 policy. Table A1 suggests that they would have been too big given their counties' patient populations. While smaller centers may have been statutorily permitted, the fixed cost of entry may have been sufficiently high to deter them.

6.4.2 Dialysis CON programs' effects on patients

Table 6 presents the 2SLS estimates and figure 4 presents the dynamic reduced-form estimates.

First, I study dialysis access outcomes. I find that marginal dialysis centers reduce distances between patients' residences and their chosen dialysis center by 8.670 miles (34% of the mean). They also reduce distances between patients' residences and their nearest dialysis center by 10.490 miles (57% of the mean). They reduce the share of patients who use HHD by 2.3 percentage points and the share who use any at-home dialysis by 7.6 percentage points.

While it is not surprising that new centers would reduce distances between patients' residences and their nearest or chosen centers, the magnitudes of the reductions are striking and consistent with letters written to the WA dialysis CON program by patients in support of potential entrants' applications to open new centers.²³ For instance, one petitioner wrote:

Last Saturday was [my husband's] 3rd dialysis treatment at Sunnyside. It's [...] so far away. [...] Most of our driving is very local. We do not leave town if there is any question of bad weather. I dread this winter. (STRESS) [...] [Each trip] takes about 7 hours in all. (3 times a week) [...] It's Monday and my husband is so dreading the long trip [...] tomorrow and saying he wants to quit and not go back. [Emphasis original]

That the marginal centers reduced at-home dialysis utilization supports the idea that dialysis CON

²³These letters were shared with me by the WA Department of Health pursuant to a public records request.

programs change patients' treatment choices: they cause some patients who intrinsically prefer in-center dialysis to choose a self-managed, at-home alternative by blocking entry.

Second, I study congestion and health outcomes. I find that marginal centers increase the stations-per-patient at an average patient's chosen center by 0.105 (46%). They also increase the nurses-per-patient at an average patient's chosen center by 0.038 (63%). These findings indicate that the marginal centers reduced congestion. But among patients with FFS Medicare as a primary payer, I do not find that marginal centers reduce monthly hospitalization rates or the share of patient-months associated with a URR below 65%, although both point estimates are negative. These results are consistent with other studies that did not find evidence of a significant association between CON programs and patient health outcomes (e.g., [Polsky et al. 2014](#); [Ho et al. 2009](#); [DiSessa et al. 2006](#); [Ho 2006](#); [Popescu et al. 2006](#)). Appendix F reports results for additional outcomes. I do not find that marginal centers reduce mortality rates, increase in-center HD patients' number of monthly HD sessions, or lower spending.

6.5 Sensitivity analysis

I conduct several sensitivity analyses to evaluate my results. See appendix G for details. First, I evaluate whether my estimates are sensitive to the control variables. Since equations (3)-(5) do not include patient fixed effects, my estimates may be sensitive to changes in the composition of the event or comparison counties' patient populations that happens to coincide with the events.²⁴ I re-estimate equation (3) without patient-level control variables. The results are qualitatively similar.

Second, I evaluate whether my estimates are sensitive to using a nationwide set of comparison counties. I re-estimate equation (3) with alternative comparison counties. For the NC threshold-crossings, I use counties in North Carolina and neighboring states (GA, VA, TN, and SC). For the 2007 WA policy change, I use counties in the Western United States (OR, ID, MT, WY, CA, NV, NM, AZ, UT, and CO). The results are qualitatively similar.

Third, I evaluate whether my estimates are sensitive to pooling WA and NC events. I re-estimate equations (3)-(5) separately for the 2007 WA policy change and the NC threshold-crossings. The results are qualitatively similar—though the sample size for WA is smaller and the first stage is weaker—suggesting that my findings are not driven by pooling the two states' events.

²⁴It is not feasible to use patient FEs because many patients do not survive throughout each event's event window.

6.6 Welfare

In section 6.4, I found that centers on the WA and NC dialysis CON programs' margins improved access, reduced congestion, and reduced at-home dialysis utilization. In this section, I measure their total contributions to patient welfare using a structural model of patient preferences. I compare the marginal centers' estimated patient welfare contributions to estimates of their reported fixed costs.

6.6.1 Patient preferences

Define the indexes i for patients, j for alternatives in the set $\mathcal{J}$, and t for time periods in the set $\mathcal{T}$. Assume that each patient's alternatives are in-center dialysis at any dialysis center within 50 miles of their residence, in-center dialysis 50 miles away, or an at-home dialysis outside option. Let u_{ijt} be patient i 's utility for being treated at alternative j in period t . Assume that:

$$u_{ijt} = \gamma \text{TrDist}_{ijt} + \lambda_j \text{Stay}_{ijt} + \delta_{jt} + \varepsilon_{ijt} \quad (6)$$

where γ measures disutility for travel, Stay_{ijt} indicates that j was i 's chosen alternative in the previous period, λ_j measures switching costs, δ_{jt} is an alternative-period fixed effect absorbing all time-varying alternative characteristics, and ε_{ijt} is an i.i.d. type 1 extreme value taste shock. Under these assumptions, each patient's expected utility in each month given their choice set is:

$$\bar{u}_{it} := \ln \left(\sum_{j \in \mathcal{J}} \exp \left(-\text{TrDist}_{ijt} + \frac{\lambda_j}{-\gamma} \text{Stay}_{ijt} + \frac{\delta_{jt}}{-\gamma} \right) \right) \quad (7)$$

I express expected utility in miles traveled-equivalents (MTEs) by scaling the parameters by $-\gamma$ because I do not observe patients making dollar-valued transactions. A one unit increase in (7) is equal to the welfare gain of reducing travel distances to all alternatives by one mile.

The model enables measuring three kinds of welfare gains associated with new centers. First, it enables measuring gains from reduced travel costs. The parameter $-\gamma$ measures the value of living closer to dialysis centers, including lower fuel costs, opportunity costs of time, and weather-related uncertainty. Second, the model enables measuring gains from a better match to in-center or at-home dialysis. The combination of γ , λ_j , and δ_{jt} measure the value of switching from at-home dialysis to

in-center dialysis when a new dialysis center opens nearby. Finally, the model enables measuring gains from a change in the mix of dialysis centers. The parameters δ_{jt} measure the value of each alternative relative to at-home dialysis, including the quality of their staff, stations, and amenities.

I estimate the model using a (patient, month, alternative)-level sample, where a patient-month is included in the sample if it appeared in the IV-DID estimation sample. Initially, there are 6 million patient-months (cases) and 4,994 alternatives, implying more than 800,000 elements of $(\lambda_j : j \in \mathcal{J})$ and $(\delta_{jt} : jt \in \mathcal{J} \times \mathcal{T})$. I simplify the computational problem by restricting $\lambda_j = \lambda$ and $\delta_{jt} = \delta_j$ for all j, t . As a result, γ is estimated from the association between conditional choice probabilities (CCPs) and travel distances, λ is estimated from differences between incident and prevalent patients' CCPs, and each δ_j is estimated from the difference between at-home dialysis utilization anywhere and in-center dialysis utilization at center j . I further reduce the computational cost by estimating the model using the event counties and the regional comparison counties described in section 6.5. The final sample consists of 1.4 million cases and 1,312 alternatives.²⁵

Figure 5 reports the estimates. Panel (a) shows that $\hat{\gamma}$ is -0.109 and $\hat{\lambda}$ is 5.875, indicating that patients dislike travel and face switching costs. The switching cost estimate is $-\hat{\lambda}/\hat{\gamma} = 53.8$ MTEs, which is high and consistent with my finding that patients switch centers in only 2% of months on average. Panel (b) plots the distribution of estimated MTE-valued alternative-specific fixed effects. It shows that 95% are positive, indicating that patients tend to prefer in-center to at-home dialysis. It also shows that the centers that opened in the event counties had above-median fixed effects, indicating that they were relatively high quality. The mean-squared choice error is 0.29%.²⁶

I use the model's estimates to compute $\hat{u}_{it}$ using the plug-in analogue of equation (7). I re-estimate equations (3)-(5) using $\hat{u}_{it}$ as an outcome. Table 6 and figure 4 report the results. They suggest that the marginal centers in the event counties produced a 10.560 MTE expected utility gain per resident patient-month on average. I also aggregate $\hat{u}_{it}$ to the (county, month)-level and re-estimate equations (3)-(5) with a stacked county-month dataset. Table 7 reports the result. The patient welfare gains associated with a center on the WA and NC dialysis CON programs' margins sum to 359 MTEs per county-month on average.

²⁵In appendix G, I add TrDist_{ijt}^2 to allow for quadratic preferences over distance. The results are qualitatively similar.

²⁶The MSCE is the average squared difference between an indicator for each case's choice and its estimated CCPs. A low MSCE indicates that estimated CCPs are close to 1 for each case's chosen alternative, and close to zero otherwise.

6.6.2 Dialysis centers' reported fixed costs

I use HCRIS data to measure centers' reported fixed costs during 2011-2016.²⁷ Since the event windows span 1992-2016, I use the available data to predict reported fixed costs prior to 2011. I estimate the model $\ln(\text{FixedCosts}_{jt}) = \beta X_{jt} + \varepsilon_{jt}$, where FixedCosts_{jt} is center j 's reported fixed cost in month t and X_{jt} is a vector of center characteristics that I observe throughout the sample period, including age, state, chain affiliation, patient volume, and stock of stations. Table A2 reports the estimates of β . It shows that there is a positive and concave association between reported fixed costs, volume, and stations. There is also heterogeneity by state, chain, and age (not shown).

I use the estimates to predict monthly fixed costs during 1992-2016 and aggregate them to the (county, month)-level. I re-estimate equations (3)-(5) with a stacked county-month dataset using the predicted values as an outcome. Table 7 reports the results. It shows that marginal centers increased reported fixed costs by \$43,568 per month on average.

6.6.3 Total welfare

The foregoing estimates contribute to a framework for evaluating the trade-off between the marginal centers' benefits for patients and their effect on costs. Since the CATE-ES is the causal effect of marginal dialysis centers in the event counties in the years following the events, I focus on the counterfactual in which the the marginal centers were delayed further.

First, table 7 shows that marginal centers increase expected utility among the residents of their county by 359 MTEs per county-month on average. This estimate reflects factors that influence patients' choices, but it is not clear that patients' choices reflect all relevant factors. For instance, marginal centers may affect health in ways that are difficult for patients to understand (Arrow 1963). While I did not find that marginal centers improve observed health outcomes, they may affect some unobserved health outcomes, or their effects may be too small to detect in this sample. The marginal centers may also benefit patients in other counties who might travel there for care.²⁸ Thus, 359 MTEs may somewhat understate the marginal centers' benefits for patients.

Second, the dollar value of an MTE is unknown. (Recall that it is not identified because I do

²⁷They are the sum of lines 1-4, 6, and 11 of Form CMS-265-11 Worksheet A, and include "expenses pertaining to buildings and fixtures such as depreciation, insurance interest, [and] rent," "expenses incurred in the operation and maintenance of the plant," and costs of "fiscal services, legal services, accounting, [and] recordkeeping," among others.

²⁸On average in a given month, 25% of patients treated at centers in the event counties lived in a non-event county.

not observe patients making transactions in dollars.) In-center dialysis patients typically make 6 one-way trips per week and there are 4.3 weeks per month on average, so the dollar value of one MTE may be equal to the dollar value of reducing monthly travel by 25.8 miles. Correspondingly, the dollar value of 359 MTEs may be equal to the dollar value of reducing monthly travel by 9,254 miles. The cost of travel may be low for patients who are relatively healthy, independent, and have a low opportunity cost of time, but it may be high for others, such as those who need a friend or relative to drive, or who cannot sit comfortably on long trips. Estimates of the resource cost of travel—such as gasoline and automobile wear—range from \$0.52-1.32 per mile.²⁹

Third, the marginal centers' effect on total costs is their fixed costs plus their effect on total variable costs. To illustrate, assume that other centers would have treated the marginal centers' patients had the marginal centers not opened, and refer to the marginal centers' patients as switchers and the other centers as incumbents. The marginal centers' effect on total variable costs is the difference between their cost of treating switchers and the incumbents' cost of treating switchers. Studies suggest that dialysis centers' variable costs are a convex function of volume because congestion raises the cost of treating marginal patients (e.g., [Eliason 2022](#)). Consequently, it is likely that the marginal centers lowered total variable costs by spreading patients across more facilities.

As for fixed costs, relative to the counterfactual that the marginal centers would have opened later, the only relevant fixed cost is the everyday cost of “keeping the lights on.” Table 7 shows that the marginal centers' estimated reported fixed costs were \$43,568 per month on average. Recall that this estimate is based on accounting cost data reported by dialysis centers to CMS through HCRIS. Studies suggest that some providers overstate their costs in these reports in part because they are used in aggregate by public payers to set prices (e.g., [Gandhi and Olenski 2024](#)). They may also indirectly include some variable costs, such as variable billing and recordkeeping costs.

In sum, I find that CON programs worsen treatment access, patients' matches with their desired treatment modalities, and congestion, through their direct effect on entry. I find that the marginal centers raise total welfare if the dollar value of 359 MTEs per month on average exceeds \$43,568; i.e., if the dollar value of an MTE is \$121 or more. This threshold is likely an upper bound, because 359 MTEs may understate their benefits for patients and \$43,568 may overstate their effect on costs.

²⁹This is equal to \$0.37-\$0.93 per driving mile ([AAA 2010](#)) times 1.417 driving miles per straight-line mile ([Boscoe et al. 2012](#)). All distances reported in this paper are straight-line miles between ZCTA centroids.

7 Conclusion

In this paper, I study how regulatory entry barriers in health care provider markets affect market structure and welfare. In particular, I study CON programs—which empower health planners to block capital investments that do not meet a community’s “needs”—in the US dialysis industry—where patients are sensitive to access and quality and provider markets are already concentrated.

I use novel sources of quasi-experimental policy variation to make three findings. First, I analyze the application process in North Carolina to find that the NC dialysis CON program reduces entry, protects incumbents, and enables incumbents to expand. Second, I leverage natural experiments in Washington and North Carolina to find that new centers on CON programs’ policy margins improve access and the patient-treatment match, and reduce congestion. I also find suggestive evidence that they improve health, but these estimates are not statistically significant. Finally, I combine a structural model of patient preferences with an IV-DID design to show that the marginal centers raise monthly countywide patient welfare by the utility value of reducing travel by 9,254 miles.

My findings are relevant to ongoing policy discussions about health system capacity, demand fluctuations, and CON reform post-COVID—including in West Virginia, Tennessee, and South Carolina (e.g., [Rice 2024](#); [Friedman 2024](#); [SC Office of the Governor 2023](#); [Hoe 2022](#); [Bernstein et al. 2020](#); [Abelson 2020](#)). Outside health economics, my findings are also related to IO studies about business-stealing externalities and excessive entry (e.g., [Seim and Waldfogel 2013](#); [Gowrisankaran and Krainer 2011](#); [Davis 2006](#); [Rysman 2004](#); [Berry and Waldfogel 1999](#); [Mankiw and Whinston 1986](#)), to which I contribute empirical evidence of a social planner’s efforts to regulate capacity.

However, further research is needed. First, that CON programs protect incumbents from potential competition may cause providers to expand faster to become the incumbents being protected instead of the potential entrants being blocked. This CON-induced preemption may improve access and raise costs along some margins. Second, CON programs may influence market structure not only through their effect on entry, but also through their effect on mergers, acquisitions, and firm turnover rates (e.g., [Wollmann 2024](#)). They may thereby delay the expansion of new, for-profit or private equity business models (e.g., [Gupta et al. 2023](#); [Eliason et al. 2020](#)). Finally, although this paper shows that CON programs protect incumbents from potential competition, their effects on dialysis centers’ prices and quality are unknown. Additional research in these areas may be fruitful.

8 Bibliography

1. AAA. 2010. "Behind the numbers: your driving costs. How much are you really paying to drive?" Available online at <https://exchange.aaa.com/wp-content/uploads/2012/04/201048935480.Driving-Costs-2010.pdf> in October 2021.
2. Abelson, Reed. 2020. "Dialysis patients face close-up risk from coronavirus." *The New York Times*. April 11, 2020.
3. Alaska Department of Health (ADOH). 2024. "Certificate of need (CON) program summary." Available online at <https://health.alaska.gov/Commissioner/Pages/RateReview/CertificateOfNeed/default.aspx>. Accessed on June 2, 2024.
4. American Health Planning Association (AHPA). 2016. "National Directory of State Certificate of Need Programs and Health Planning Agencies."
5. American Health Planning Association (AHPA). 2006. "Health Services Planning and CON Regulation in Mississippi."
6. Arrow, Kenneth J. 1963. "Uncertainty and the welfare economics of medical care." *The American Economic Review* 53(5): 1963.
7. Baker, Matthew C. and Thomas Stratmann. 2021. "Barriers to entry in the healthcare markets: winners and losers from certificate-of-need laws." *Socio-economic planning sciences* 77.
8. Bernstein, Lenny, Christopher Rowland, and Tom Hamburger. 2020. "Dialysis patients are at high risk during covid-19 outbreak." *The Washington Post*. March 24, 2020.
9. Berry, Steven T., and Joel Waldfogel. 1999. "Free entry and social inefficiency in radio broadcasting." *The RAND Journal of Economics* 30(3).
10. Bertuzzi, Luca, Paul J. Eliason, Benjamin Heebsh, Riley J. League, Ryan C. McDevitt, and James W. Roberts. 2023. "Gaming and effort in performance pay." NBER Working Paper 31353.
11. Blumstein, James F. and Frank A. Sloan. 1978. "Health Planning and Regulation Through Certificate of Need: An Overview." *Utah Law Review* 3: 3-38.
12. Boscoe, Francis P., Kevin A. Henry, and Michael S. Zdeb. 2012. "A nationwide comparison of driving distance versus straight-line distance to hospitals." *The Professional Geographer* 64 (2).
13. Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik. 2014a. "Robust nonparametric confidence intervals for regression discontinuity designs." *Econometrica* 82(6).
14. Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik. 2014b. "Robust data-drive inference in the regression discontinuity design." *The Stata Journal* 14(4).
15. Campbell, Ellen S. and Gary M. Fournier. "Certificate-of-need deregulation and indigent hospital care." *Journal of Health, Politics, and Law* 18(4): 905-25.
16. Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The effect of minimum wages on low-wage jobs." *The Quarterly Journal of Economics* 134(3): 1405-54.

17. Centers for Medicare and Medicaid Services (CMS). 2019. “Medicare benefit policy manual chapter 11: end stage renal disease.” Available online at <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/Downloads/bp102c11.pdf>. Accessed in July 2024.
18. Centers for Medicare and Medicaid Services (CMS). 2023. “Medicare secondary payer (MSP) end stage renal disease (ESRD).” Available online at <https://www.cms.gov/Medicare/Coordination-of-Benefits-and-Recovery/Coordination-of-Benefits-and-Recovery-Overview/Medicare-Secondary-Payer/Downloads/MSP-End-Stage-Renal-Disease-ESRD.pdf>. Accessed in July 2024.
19. Centers for Medicare and Medicaid Services (CMS). 2024a. “National Health Expenditures Summary, including share of GDP, CY 1960-2022.” Available online at <https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical>. Accessed on June 1, 2024.
20. Centers for Medicare and Medicaid Services (CMS). 2024b. “Medicare coverage of kidney dialysis & kidney transplant services.” Available online at <https://www.medicare.gov/publications/10128-medicare-coverage-esrd.pdf>. Accessed in July 2024.
21. Childers, Christopher P., Jill Q. Dworsky, Gerald Kominski, and Melinda Maggard-Gibbons. 2019. “A comparison of payments to a for-profit dialysis firm from government and commercial insurers.” *JAMA Internal Medicine* 179(8).
22. Cutler, David M., Robert S. Huckman, and Jonathan T. Kolstad. 2010. “Input constraints and the efficiency of entry: lessons from cardiac surgery.” *The American Economic Journal: Economic Policy* 2(1).
23. Dai, Mian. 2014. “Product choice under price regulation: evidence from out-patient dialysis markets.” *International Journal of Industrial Organization* 32: 24-32.
24. Dai, Mian and Xun Tang. 2015. “Regulation and capacity competition in health care: evidence from U.S. dialysis markets.” *The Review of Economics and Statistics* 97(5): 965-82.
25. Davis, Peter. 2006. “Measuring the business stealing, cannibalization, and market expansion effects of entry in the US motion picture exhibition market.” *The Journal of Industrial Economics* 54(3).
26. DiSesa, Verdi J., Sean M. O’Brien, Karl F. Welke, Sarah M. Beland, Constance K. Haan, Mary S. Vaughan-Sarrazin, and Eric D. Peterson. 2006. “Contemporary impact of state certificate-of-need regulations for cardiac surgery. An analysis using the society of thoracic surgeons’ national cardiac surgery database.” *Circulation* 114: 2122-29.
27. Department of Justice and Federal Trade Commission (DOJ and FTC). 2004. “Improving Health Care: A Dose of Competition.” July 2004.
28. Department of Justice and Federal Trade Commission (DOJ and FTC). 2008. “Joint statement of the antitrust division of the U.S. Department of Justice and the Federal Trade Commission Before the Illinois Task Force on Health Planning Reform.” Available on https://www.justice.gov/archive/atr/public/press_releases/2008/237153a.pdf. Accessed on June 2, 2024.

29. Duflo, Esther. 2001. "Schooling and labor market consequences of school construction in Indonesia: evidence from an unusual policy experiment." *American Economic Review* 91 (4): 795-813.
30. Eastaugh, Steven T. 1982. "The effectiveness of community-based hospital planning: some recent evidence." *Applied Economics* 14: 475-90.
31. Eliason, Paul J. 2022. "Price regulation and market structure: evidence from the dialysis industry." Working paper available at <https://sites.google.com/view/pauljeliason/research>.
32. Eliason, Paul J., Benjamin Heebsh, Ryan C. McDevitt, and James W. Roberts. 2020. "How acquisitions affect firm behavior and performance: evidence from the dialysis industry." *The Quarterly Journal of Economics* 135(1): 221-67.
33. Eliason, Paul J., Benjamin Heebsh, Riley J. League, Ryan C. McDevitt, and James W. Roberts. 2022. "The effect of bundled payments on provider behavior and patient outcomes." Working paper dated February 2022.
34. Ferguson, Thomas W., Reid H. Whitlock, Ryan J. Bamforth, Alain Beaudry, Joseph Darcel, Michelle Di Nella, Claudio Rigatto, Navdeep Tangri, and Paul Komenda. 2021. "Cost-utility of dialysis in Canada: hemodialysis, peritoneal dialysis, and nondialysis treatment of kidney failure." *Kidney Medicine* 3(1): 20-30.
35. Ford, Jon M. and David L. Kaserman. 1993. "Certificate-of-need regulation and entry: evidence from the dialysis industry." *Southern Economic Journal* 59(4): 783-91.
36. Friedman, Adam. 2024. "Tennessee passes legislation to reform hospital certificate of need law." *Tennessee Lookout*. April 25, 2024.
37. Gandhi, Ashvin, and Andrew Olenski. 2024. "Tunneling and hidden profits in health care." NBER Working Paper 32258.
38. Gardner, John, Neil Thakral, Linh T. To, and Luther Yap. 2024. "Two-stage difference in differences." Working paper available at https://jrgcmu.github.io/2sdd_gtty.pdf.
39. Goodman-Bacon, Andrew. 2021. "Difference-in-differences with variation in treatment timing." *Journal of Econometrics* 225(2): 254-77.
40. Gowrisankaran, Gautam, and John Krainer. 2011. "Entry and pricing in a differentiated products industry: evidence from the ATM market." *The RAND Journal of Economics* 42(1).
41. Government Accountability Office (GAO). 2015. Medicare payment refinements could promote increased use of home dialysis. GAO Report 16-125.
42. Grieco, Paul L.E. and Ryan C. McDevitt. 2017. "Productivity and quality in health care: evidence from the dialysis industry." *Review of Economic Studies* 84: 1071-1105.
43. Gupta, Atul, Sabrina T. Howell, Constantine Yannelis, and Abhinav Gupta. 2023. "Owner incentives and performance in healthcare: private equity investment in nursing homes." *The Review of Financial Studies* 37 (4): 1029-77.
44. Hellinger, Fred J. 1976. "The effect of certificate-of-need legislation on hospital investment." *Inquiry* 13(2): 187-93.
45. Ho, Vivian. 2006. "Does certificate of need affect cardiac outcomes and costs?" *International Journal of Health Care Finance and Economics* 6: 300-24.

46. Ho, Vivian, Meei-Hsiang Ku-Goto, and James G. Jollis. 2009. "Certificate of need (CON) for cardiac care: controversy over the contributions of CON." *HSR: Health Services Research* 44(2): 483-500.
47. Hoe, Thomas. 2022. "Does hospital crowding matter? Evidence from trauma and orthopedics in England." *American Economic Journal: Economic Policy* 14(2): 231-62.
48. Horwitz, Jill, Austin Nichols, Carrie H. Colla, and David M. Cutler. 2024. "Technology regulation reconsidered: the effects of certificate of need policies on the quantity and quality of diagnostic imaging." NBER Working Paper 32143.
49. Klomjit, Nattawat, Andrea G. Kattah, and Wisit Cheungpasitporn. 2021. "The cost-effectiveness of peritoneal dialysis is superior to hemodialysis: updated evidence from a more precise model." *Kidney Medicine* 3(1): 15-7.
50. Knuepfer, Jack. 1974. Transcript of the June 6, 1974 meeting of the 78th General Assembly of the State of Illinois.
51. Koopman, Christopher and Anne Philpot. 2016. "The state of certificate-of-need laws in 2016." Mercatus Center. Available online at mercatus.org/publications. Accessed on December 4, 2020.
52. Lanning, J.A., M.A. Morrissey, and R.L. Ohsfeldt. 1991. "Endogenous hospital regulation and its effects on hospital and non-hospital expenditures." *Journal of Regulatory Economics* 3: 137-54.
53. League, Riley J., Paul Eliason, Ryan C. McDevitt, James W. Roberts, and Heather Wong. 2022. "Variability in prices paid for hemodialysis by employer-sponsored insurance in the US from 2012 to 2019." *JAMA Network Open* 5(2).
54. Leavitt, Michael O. 2008. "A design for a bundled end stage renal disease prospective payment system." *Report to Congress*. U.S. Department of Health and Human Services. Washington, DC.
55. Lee, David S., Justin McCrary, Marcelo J. Moreira, Jack Porter, and Luther Yap. 2023. "What to do when you can't use 1.96 confidence intervals for IV." NBER Working Paper 31893.
56. Lin, Eugene, Bich Ly, Erin Duffy, and Erin Trish. 2022. "Medicare Advantage plans pay large markups to consolidated dialysis organizations." *Health Affairs* 41 (8): 1107-16.
57. Mankiw, Gregory N., and Michael D. Whinston. 1986. "Free entry and social inefficiency." *The RAND Journal of Economics* 17(1).
58. Mayo Clinic. 2024a. "Peritoneal dialysis." Available online at <https://www.mayoclinic.org/tests-procedures/peritoneal-dialysis/about/pac-20384725>. Accessed in July 2024.
59. Mayo Clinic. 2024b. "Kidney transplant." Available online at <https://www.mayoclinic.org/tests-procedures/kidney-transplant/about/pac-20384777> in July 2024.
60. MedPAC. 2024. *Report to Congress*. Medicare Payment Advisory Commission (MedPAC). Washington, DC, March.

61. MedPAC. 2005. *Report to Congress*. Medicare Payment Advisory Commission (MedPAC). Washington, DC, March.
62. MedPAC. 2000. *Report to Congress*. Medicare Payment Advisory Commission (MedPAC). Washington, DC, March.
63. Morgan, Paulette C., and Suzanne M. Kirchhoff. 2021. "Medicare Advantage (MA) coverage of end stage renal disease (ESRD) and network requirement changes." Congressional Research Services report R46655.
64. National Conference of State Legislatures (NCSL). 2024. "CON State list." Available at ncsl.org. Accessed in June 2024.
65. National Kidney Foundation (NKF). 2024a. "Kidney failure." Available online at <https://www.kidney.org/atoz/content/kidney-failure>. Accessed in July 2024.
66. National Kidney Foundation (NKF). 2024b. "Dialysis: deciding to stop." Available online at <https://www.kidney.org/atoz/content/dialysisstop>. Accessed in July 2024.
67. National Kidney and Urologic Diseases Information Clearinghouse (NKUDIC). 2009. "Hemodialysis dose and adequacy." Available online at https://www.niddk.nih.gov/-/media/Files/Kidney-Disease/hemodialysis_dose_508.pdf in July 2024.
68. Ohlhausen, Maureen. 2015. "Certificate of Need Laws: A Prescription for Higher Costs." *Antitrust* 30(1): 50-4.
69. Perry, Bryan J. 2017. "Certificate of need regulation and hospital behavior: evidence from MRIs in North Carolina." Working paper dated November 1, 2017.
70. Polsky, Daniel, Guy David, Jianing Yang, Bruce Kinosian, and Rachel Werner. 2014. "The effect of entry regulation in the health care sector: the case of home health." *Journal of Public Economics* 110: 1-14.
71. Popescu, Ioana, Mary S. Vaughan-Sarrazin, and Gary E. Rosenthal. 2006. "Certificate of need regulations and use of coronary revascularization after acute myocardial infarction." *JAMA* 295(18): 2141-47.
72. Posner, Richard A. 1971. "Taxation by regulation." *The Bell Journal of Economics and Management Science* 2(1): 22-50.
73. Rice, Emily. "House committee votes to eliminate healthcare certificate of need." *WV Public Broadcasting*. February 21, 2024.
74. Roemer, Milton. 1961. "Bed supply and hospital utilization. A natural experiment." *Hospitals: the Journal of the American Hospital Association* 35(1).
75. Russell, Louise B. 1979. "Regulating the diffusion of hospital technologies." *Law and Contemporary Problems* 43(1): 26-42.
76. Rysman, Marc. 2004. "Competition between networks: a study of the market for yellow pages." *The Review of Economic Studies* 71(2).
77. Salkever, David S. 2000. "Regulation of prices and investment in hospitals in the United States." *Handbook of Health Economics*, vol. 1. Editors A.J. Culyer and J.P. Newhouse. Elsevier Science B.V. 2000.

78. Salkever, David S. and Thomas W. Bice. 1976. "The impact of certificate-of-need controls on hospital investment." *The Millbank Memorial Fund Quarterly. Health and Society* 54(2): 185-214.
79. Salkever, David S. and Thomas W. Bice. 1979. "Hospital certificate-of-need controls: impact on investment, costs, and use." Washington D.C.: American Enterprise Institute for Public Policy Research.
80. Seim, Katja, and Joel Waldfogel. 2013. "Public monopoly and economic efficiency: evidence from the Pennsylvania liquor control board's entry decisions." *American Economic Review* 103(2).
81. Sloan, Frank A. 1981. "Regulation and the rising cost of hospital care." *The Review of Economics and Statistics* 63(4): 479-87.
82. Sloan, Frank A. and Bruce Steinwald. 1980. "Effects of regulation on hospital costs and input use." *The Journal of Law and Economics* 23(1): 81-109.
83. SC Office of the Governor. 2023. "Governor Henry McMaster signs certificate of need repeal." October 3, 2023. Available online at <https://governor.sc.gov/news/2023-10/governor-henry-mcmaster-signs-certificate-need-repeal>.
84. Swaminathan, Shailender, Vincent Mor, Rajnish Mehrotra, and Amal Trivedi. 2012. "Medicare's payment strategy for end-stage renal disease now embraces bundled payment and pay-for-performance to cut costs." *Health Affairs (Millwood)* 31(9): 2051-8.
85. Trish, Erin, Matthew Fiedler, Ning Ning, Laura Gascue, Loren Adler, and Eugene Lin. 2021. "Payment for dialysis services in the individual market." *JAMA Internal Medicine* 181(5): 698-9.
86. U.S. Renal Data System (USRDS). 2018. Researcher's Guide to the USRDS Database.
87. U.S. Renal Data System (USRDS). 2020. *USRDS annual data report: epidemiology of kidney disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2020.
88. U.S. Renal Data System (USRDS). 2022. *USRDS annual data report: epidemiology of kidney disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2022.
89. Washington State Register. 2006. Issue 06-24. Subpart 050. Filed December 1, 2006.
90. Wendling, Wayne and Jack Werner. 1980. "Nonprofit firms and the economic theory of regulation." *Quarterly Review of Economics and Business* 20(3).
91. Wollmann, Thomas G. 2024. "How to get away with merger: stealth consolidation and its effects on US dialysis." Working paper available online at https://faculty.chicagobooth.edu/~media/faculty/Thomas-Wollmann/Research/%20how_to_get_away_with_merger_2024.pdf.
92. Wu, Bingxiao, Jeah Jung, Hyunjee Kim, and Daniel Polsky. 2019. "Entry regulation and the effect of public reporting: evidence from home health compare." *Health Economics* 28: 492-516.

	National averages					Border county averages			
	Overall	CON?		Differences		CON?		Differences	
		No	Yes	No FE	FE _t	No	Yes	FE _e	FE _{et}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Capacity and resident patient population									
1[Has Center] ¹	0.55	0.53	0.65	0.11*	0.11*	0.72	0.63	-0.10***	-0.07
#[Centers] ¹	2.10	2.09	2.15	0.06	-0.33***	2.91	3.15	-0.00	-0.34*
#[Stations] ¹	34.79	34.29	37.06	2.77	-4.52**	49.62	53.57	-1.35	-5.04**
#[Patients] ¹	127	122	152	30	-	164	235	47	-
Facility ownership									
#[Chain-owned] ¹	0.67	0.66	0.69	0.02	-0.01	0.95	0.76	-0.20***	-0.10
#[Independent] ¹	0.69	0.66	0.84	0.18	0.03	0.83	1.32	0.40	-0.11
#[Owners] ¹	1.36	1.32	1.53	0.20	0.03	1.77	2.08	0.19	-0.21
Sample composition									
#[County-months] (M)	0.49	0.40	0.09	0.49	0.49	0.07	0.03	0.10	0.10
#[Counties]	3,144	2,572	572	3,144	3,144	218	196	414	414

Tab. 1. *Counties' dialysis patient and center characteristics during 2004-2016.* This table reports descriptive statistics comparing counties in CON and non-CON states statewide (cols. 2-5) and at border counties (cols. 6-9). In columns 5 and 9, the time fixed effects are interacted with the county's resident patient population. *p*-values are computed using state-level cluster-robust standard errors. * *p* < 0.10, ** *p* < 0.05, *** *p* < 0.01. See the discussions near pages 8 and 10. Source: author's analysis of the USRDS Database.

¹ Estimated from a (county, month)-level sample.

	National averages					Border county averages			
	Overall (1)	CON?		Differences		CON?		Differences	
		No (2)	Yes (3)	No FE (4)	FE _t (5)	No (6)	Yes (7)	FE _e (8)	FE _{et} (9)
Age, race, ethnicity, and sex									
Age ¹	61.35	61.36	61.34	-0.01	-0.00	62.00	61.67	-0.65	0.01
1[White] ²	0.65	0.67	0.56	-0.11**	-0.10**	0.61	0.55	-0.06	0.03
1[Black] ²	0.29	0.28	0.37	0.09	0.09	0.36	0.40	0.06	-0.02
1[AIAN] ²	0.01	0.01	0.01	-0.01	-0.01	0.00	0.01	0.00	-0.00
1[Asian] ²	0.03	0.03	0.04	0.01	0.01	0.02	0.04	0.00	-0.01
1[NHPI] ²	0.01	0.01	0.01	0.01	0.01	0.01	0.00	-0.00	-0.00
1[Hispanic] ²	0.14	0.15	0.08	-0.07	-0.07	0.06	0.12	0.02	-0.01***
1[Not Hispanic] ²	0.86	0.84	0.91	0.07	0.07	0.94	0.87	-0.02	0.01**
1[Female] ²	0.44	0.44	0.44	0.01	0.01	0.44	0.44	0.00	0.00
1[Male] ²	0.56	0.56	0.56	-0.01	-0.00	0.56	0.56	-0.00	-0.00
Employment status at ESRD onset									
1[Some empl.] ²	0.11	0.11	0.11	0.00	0.00	0.12	0.11	-0.01*	-0.02**
1[No empl.] ²	0.84	0.84	0.84	-0.00	-0.00	0.83	0.84	0.00	0.02***
Height and weight at ESRD onset									
Height (cm) ²	168.14	168.13	168.19	0.05	0.06	168.64	167.89	-0.12	0.22
Weight (kg) ²	82.06	82.00	82.27	0.27	0.31	82.78	80.99	-0.37	-0.57
BMI ²	28.96	28.94	29.03	0.09	0.09	29.07	28.68	-0.07	-0.25**
Primary cause of ESRD									
1[Diabetes] ²	0.44	0.44	0.43	-0.02	-0.02	0.41	0.41	0.00	-0.01
1[Hypertension] ²	0.28	0.28	0.29	0.00	0.00	0.32	0.30	-0.01	0.00
1[Glomer.] ²	0.10	0.10	0.10	0.01	0.00	0.10	0.10	-0.00	-0.00
1[Cyst. Kidney] ²	0.02	0.02	0.02	0.00	-0.00	0.02	0.02	-0.00	-0.00
Sample composition									
#[Patient-months] (M)	62.98	49.32	13.56	62.88	62.88	10.72	7.17	17.89	17.89
#[Patients] (M)	1.79	1.41	0.38	1.79	1.79	0.17	0.21	0.38	0.38

Tab. 2. *Patients' pre-dialysis characteristics during 2004-2016.* This table reports descriptive statistics comparing patients in CON and non-CON states statewide (cols. 2-5) and at border counties (cols. 6-9). In columns 5 and 9, the time fixed effects are interacted with each patient's county's resident patient population. *p*-values are computed using state-level cluster-robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussions near pages 8 and 10. Source: author's analysis of the USRDS Database.

¹ Estimated from a (patient, month)-level sample.

² Estimated from a patient-level sample.

	National averages					Border county averages			
	Overall	CON?		Differences		CON?		Differences	
		No	Yes	No FE	FE _t	No	Yes	FE _e	FE _{et}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Miles to nearest alternatives									
Any alternative ¹	3.73	3.75	3.66	-0.09	-0.16	3.19	2.43	0.19	0.24
≥50% Stats/Pats ¹	7.80	7.63	8.43	0.79	0.67	5.85	5.06	0.50	1.37*
Alternatives within 100 miles									
#[Centers] ¹	161	155	185	30.08	31.44	202	278	1.84	-7.45
#[Chains] ¹	5.23	5.27	5.12	-0.15	-0.14	5.80	6.46	-0.07	-0.18***
#[Independent] ¹	49.25	44.70	65.81	21.11	21.27	70.39	106.39	1.18	-3.67
#[Owners] ¹	54.48	49.96	70.93	20.96	21.13	76.18	112.85	1.12	-3.85
Chosen center characteristics									
Distance (mi) ¹	10.47	10.66	9.77	-0.89	-0.97	9.16	8.21	0.31	0.02
#[Stats/Pats] ¹	0.25	0.25	0.23	-0.02	-0.02	0.26	0.22	-0.03***	-0.03***
#[Nurses/Pats] ¹	0.07	0.07	0.08	0.00	0.00	0.08	0.08	-0.00	-0.00
#[PCTs/Pats] ¹	0.08	0.08	0.08	0.00	0.00	0.08	0.09	0.00	0.00
#[Dietic./Pats] ¹	0.01	0.01	0.01	-0.00	-0.00	0.01	0.01	-0.00	-0.00*
1[Not switched] ¹	0.98	0.98	0.98	-0.00	-0.00	0.98	0.98	-0.00	0.00
1[Switched] ¹	0.02	0.02	0.02	0.00	0.00	0.02	0.02	0.00	-0.00
Treatment modalities									
1[In-center HD] ¹	0.90	0.90	0.90	0.00	0.00	0.90	0.92	0.01	0.01
1[PD] ¹	0.09	0.09	0.08	-0.01	-0.01	0.09	0.06	-0.01	-0.01
1[HHD] ¹	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.00*	0.00
1[Ever TX] ²	0.14	0.14	0.14	0.00	-0.00	0.14	0.14	-0.00	-0.02
Sample composition									
#[Patient-months] (M)	62.98	49.32	13.56	62.88	62.88	10.72	7.17	17.89	17.89
#[Patients] (M)	1.79	1.41	0.38	1.79	1.79	0.17	0.21	0.38	0.38

Tab. 3. *Patients' treatment choices during 2004-2016.* This table reports descriptive statistics comparing patients in CON and non-CON states statewide (cols. 2-5) and at border counties (cols. 6-9). In columns 5 and 9, the time fixed effects are interacted with each patient's county's resident patient population. *p*-values are computed using state-level cluster-robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussions near pages 8 and 10. Source: author's analysis of the USRDS Database.

¹ Estimated from a (patient, month)-level sample.

² Estimated from a patient-level sample.

	National averages					Border county averages			
	Overall	CON?		Differences		CON?		Differences	
		No	Yes	No FE	FE _t	No	Yes	FE _e	FE _{et}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Health insurance enrollment and spending									
1[MCare FFS Prim.] ¹	0.70	0.69	0.71	0.02	0.02	0.74	0.69	-0.04**	-0.00
1[HMO] ¹	0.11	0.12	0.09	-0.03	-0.03	0.08	0.09	0.02	-0.00
1[Any Medicare] ¹	0.78	0.78	0.79	0.02	0.01	0.83	0.77	-0.05**	-0.01
1[Dual Eligible] ¹	0.36	0.37	0.35	-0.01	-0.01	0.33	0.34	0.00	0.04
Medicare spending ^{1,2}	5,313	5,320	5,288	-32.06	-19.10	5,424	5,699	1.37	-72.68
Dialysis sessions									
#[In-center HD] ^{1,2}	10.70	10.70	10.71	0.01	0.00	10.81	10.80	-0.10	-0.03
#[HHD] ^{1,2}	0.20	0.18	0.28	0.09	0.09	0.19	0.27	0.06*	0.02
#[PD] ^{1,2}	1.49	1.50	1.46	-0.04	-0.06	1.55	1.15	-0.22	-0.17
Urea reduction ratio (URR)									
1[URR <65%] ^{1,2}	0.08	0.08	0.08	0.00	-0.00	0.07	0.08	0.01**	0.00
1[URR 65-75%] ^{1,2}	0.44	0.44	0.43	-0.02	-0.02	0.46	0.44	-0.01	-0.00
1[URR >75%] ^{1,2}	0.47	0.47	0.49	0.02	0.02	0.46	0.48	-0.00	0.00
Hospitalizations and mortality									
1[Any] ^{1,2}	0.17	0.17	0.16	-0.00	-0.00	0.17	0.17	-0.00	0.00*
1[Circulatory] ^{1,2}	0.06	0.06	0.06	-0.00	-0.00	0.06	0.06	-0.00	0.00
1[Kidney and UT] ^{1,2}	0.02	0.02	0.02	-0.00	-0.00	0.02	0.02	0.00	0.00***
1[Infections] ^{1,2}	0.01	0.01	0.01	-0.00	-0.00	0.02	0.02	-0.00	0.00
1[Died] ¹	0.02	0.02	0.02	-0.00	-0.00	0.02	0.02	-0.00	0.00
Sample composition									
#[Patient-months] (M)	62.98	49.32	13.56	62.88	62.88	10.72	7.17	17.89	17.89
#[Patients] (M)	1.79	1.41	0.38	1.79	1.79	0.17	0.21	0.38	0.38

Tab. 4. *Patients' health outcomes and spending during 2004-2016.* This table reports descriptive statistics comparing patients in CON and non-CON states statewide (cols. 2-5) and at border counties (cols. 6-9). In columns 5 and 9, the time fixed effects are interacted with each patient's county's resident patient population. *p*-values are computed using state-level cluster-robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussions near pages 8 and 10. Source: author's analysis of the USRDS Database.

¹ Estimated from a (patient, month)-level sample.

² Sub-sample associated with Medicare FFS as a primary payer.

	National averages					Border county averages			
	Overall	CON?		Differences		CON?		Differences	
		No	Yes	No FE	FE _t	No	Yes	FE _e	FE _{et}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
Age and chain affiliation									
Age ¹	16.62	16.32	17.90	1.59**	1.60**	16.13	19.08	1.48*	2.39
1[Davita] ²	0.28	0.29	0.21	-0.08**	-0.08**	0.29	0.23	-0.04	-0.04
1[Fresenius] ²	0.29	0.28	0.32	0.04	0.04	0.32	0.29	-0.04	-0.04
1[Small chain] ²	0.14	0.15	0.11	-0.04**	-0.04**	0.15	0.10	-0.04***	-0.00
1[Independent] ²	0.29	0.28	0.36	0.08	0.08	0.24	0.39	0.12***	0.08
Profit status and hospital affiliation									
1[For-profit] ²	0.79	0.80	0.73	-0.07	-0.07	0.81	0.73	-0.07**	-0.12*
1[Not for-profit] ²	0.21	0.20	0.27	0.07	0.07	0.19	0.27	0.07**	0.12*
1[Freestanding] ²	0.86	0.87	0.82	-0.05	-0.05	0.88	0.79	-0.06**	-0.09
1[Hosp. based] ²	0.14	0.13	0.18	0.05	0.05	0.12	0.21	0.06**	0.09
Monthly patient volume									
#[Patients] ¹	69.80	67.77	78.48	10.71*	10.48*	65.64	81.56	6.64*	8.43**
#[In-center HD] ¹	62.83	60.96	70.83	9.87*	9.67	59.06	75.27	7.27**	8.55**
#[HHD] ¹	0.96	0.84	1.47	0.63	0.65*	0.76	1.44	0.27	-0.10
#[PD] ¹	6.18	6.14	6.38	0.25	0.20	5.98	5.02	-0.92	0.01
Capacity and monthly congestion									
#[Stations] ²	17.05	16.91	17.68	0.77	0.74	17.22	17.47	-0.52	-0.64
#[Stats/Pats] ¹	0.77	0.76	0.81	0.05	0.05	0.76	1.04	0.22*	0.10
#[Nurses/Pats] ¹	0.16	0.14	0.23	0.10	0.09	0.16	0.28	0.11***	0.02
#[PCTs/Pats] ¹	0.13	0.12	0.16	0.03	0.03	0.12	0.19	0.06**	-0.05
#[Dietic./Pats] ¹	0.03	0.02	0.03	0.01	0.01	0.02	0.04	0.01**	-0.00
Miles to nearest center									
To Co-owned ¹	12.76	12.65	13.28	0.63	0.68	10.48	9.45	-0.27	2.10
To Competing ¹	8.29	8.00	9.57	1.57	1.60	6.54	5.47	0.39	1.29
Monthly reported costs (\$K)									
Total ^{1,3}	225	219	252	33	34	213	268	30**	45***
Capital ^{1,3}	37	36	42	6	6	37	45	4*	6**
Staff ^{1,3}	70	67	79	12	12	64	86	12**	16**
Supply ^{1,3}	24	23	28	5*	5*	22	28	4**	5**
Admin ^{1,3}	55	54	59	5	6	52	62	4	10**
Other ^{1,3}	40	39	45	6	6	37	48	6**	9***
Fixed ^{1,3}	92	90	101	11	11	89	106	7*	16**
Average ^{1,3}	3.4	3.4	3.4	0.0	0.0	3.5	3.4	-0.1	-0.3***
Sample composition									
#[Centers]	8,336	6,801	1,535	8,336	8,336	774	769	1,543	1,543
#[Center-years] (K)	79	64	15	79	79	7	7	15	15
#[Center-months] (M)	0.91	0.74	0.17	0.91	0.91	0.17	0.09	0.25	0.25

Tab. 5. *Centers' characteristics during 2004-2016.* This table reports descriptive statistics comparing dialysis centers in CON and non-CON states statewide (cols. 2-5) and at border counties (cols. 6-9). In columns 5 and 9, the time fixed effects are interacted with each center's county's resident patient population. p -values are computed using state-level cluster-robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussions near pages 8 and 10. Source: author's analysis of the USRDS Database and HCRIS data.

¹ Estimated from a (center, month)-level sample.

² Estimated from a (center, year)-level sample.

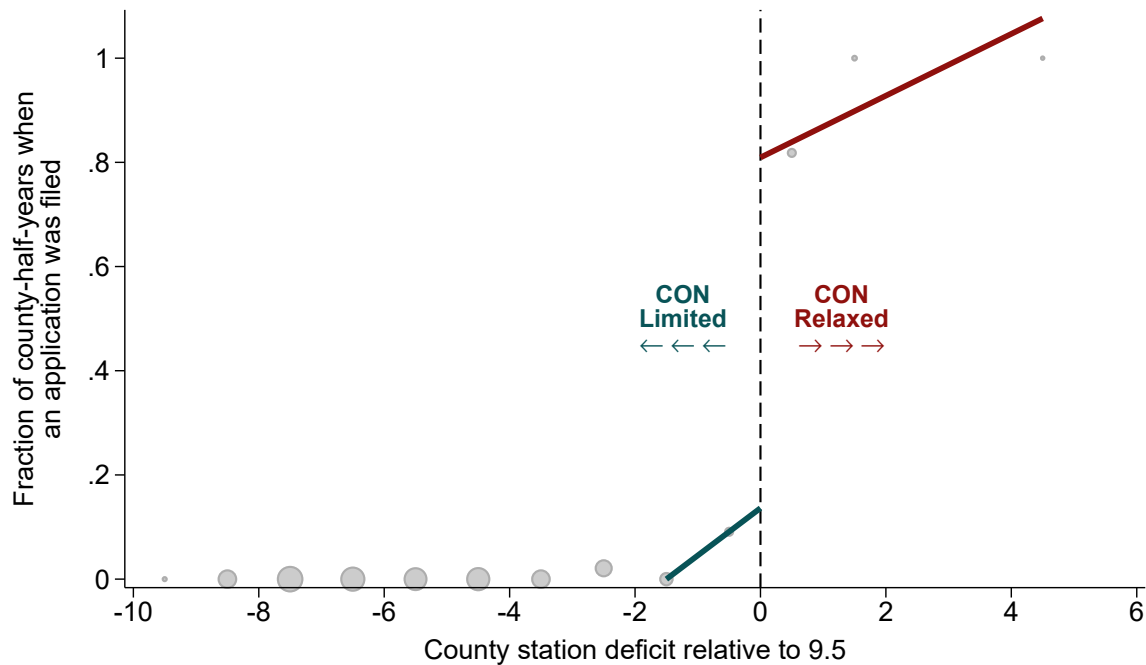
³ Available only during 2011-2016.

	Mi. to Chosen Center (1)	Mi. to Nearest Center (2)	1[HHD] (3)	1[PD] (4)	1[Home] (5)
1[Has Center]	-8.670*** (1.565) [5.538]	-10.490*** (1.860) [5.640]	-0.023** (0.011) [2.147]	-0.053 (0.037) [1.461]	-0.076** (0.036) [2.099]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	USA	USA	USA	USA	USA
Baseline Y	25.755	18.244	0.015	0.162	0.176
Reduced form	-4.691***	-5.590***	-0.012**	-0.028	-0.041**
First stage	0.541***	0.533***	0.533***	0.533***	0.533***
F-statistic	95.317	87.538	87.538	87.538	87.538
VtF 95% CV	1.918	1.916	1.919	1.921	1.921
#[Clusters]	1,743	1,743	1,744	1,744	1,744
#[Treated]	78,797	80,271	80,271	80,271	80,271
#[Untreated]	5,867,165	5,988,933	5,989,062	5,989,062	5,989,062
N	65,847,118	67,383,305	67,385,106	67,385,106	67,385,106
	Stats/Pats At Chosen Center (6)	Nurs./Pats At Chosen Center (7)	1[Hosp.] (8)	1[URR <65%] (9)	Expected Utility (MTEs) (10)
1[Has Center]	0.105*** (0.020) [5.215]	0.038*** (0.007) [5.668]	-0.019 (0.013) [1.530]	-0.007 (0.024) [0.295]	10.560*** (0.853) [12.383]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	USA	USA	USA	USA	Reg.
Baseline Y	0.227	0.060	0.182	0.101	54.122
Reduced form	0.057***	0.019***	-0.010	-0.003	6.028***
First stage	0.541***	0.500***	0.526***	0.447***	0.571***
F-statistic	95.318	43.730	79.717	30.832	84.692
VtF 95% CV	1.926	1.893	1.915	1.849	1.910
#[Clusters]	1,744	1,420	1,743	1,510	444
#[Treated]	78,797	31,496	66,131	36,536	78,797
#[Untreated]	5,867,331	2,874,368	4,849,749	2,883,666	1,276,451
N	65,849,227	33,349,389	54,686,612	34,143,752	8,453,167

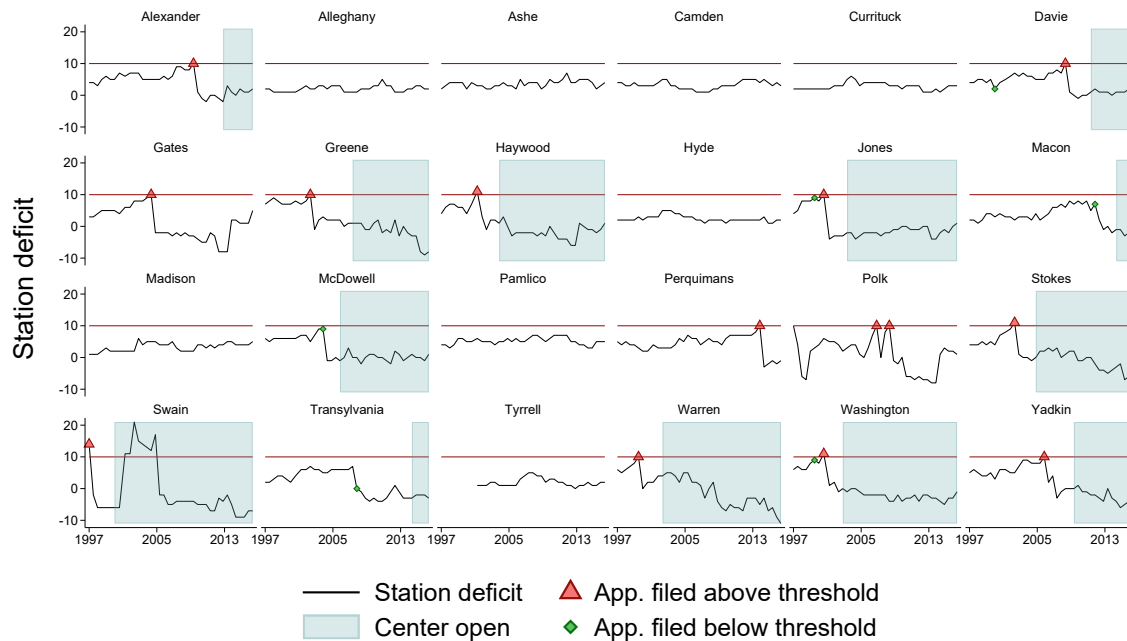
Tab. 6. *IV-DID estimates of marginal centers' effects on patient access, health, and welfare.* This table reports the main results from the IV-DID analysis. The top panel reports estimates of β in equation (3a). County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). Comparison groups are nationwide or regional counties without centers on the events' effective dates. #[Treated] and #[Untreated] refer to unique patient-months and N refers to all observations in the stacked dataset. See the discussions near pages 16 and 19. Source: author's analysis of the USRDS Database.

	Expected Utility (MTEs) (1)	Reported Fixed Costs (3)
1[Has Center]	358.680*** (56.696) [6.326]	43568.129*** (3820.682) [11.403]
County FE	Y	Y
Month FE	Y	Y
Comp. group	Reg.	Reg.
Baseline Y	974.398	0.000
Reduced form	131.679***	15,994.833***
First stage	0.367***	0.367***
F-statistic	24.121	24.121
VtF 95% CV	2.152	2.330
#[Clusters]	449	449
#[Treated]	3,744	3,744
#[Untreated]	87,166	87,166
N	729,876	729,876

Tab. 7. *IV-DID estimates of marginal centers' effects on countywide patient welfare and fixed costs.* This table reports estimates of β in equation (3a) for the countywide expected utility and fixed cost estimates. County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Comparison groups are regional counties without centers on the events' effective dates. #[Treated] and #[Untreated] refer to unique county-months and N refers to all observations in the stacked dataset. See the discussions near pages 19 and 20. Source: author's analysis of the USRDS Database and HCRIS data.

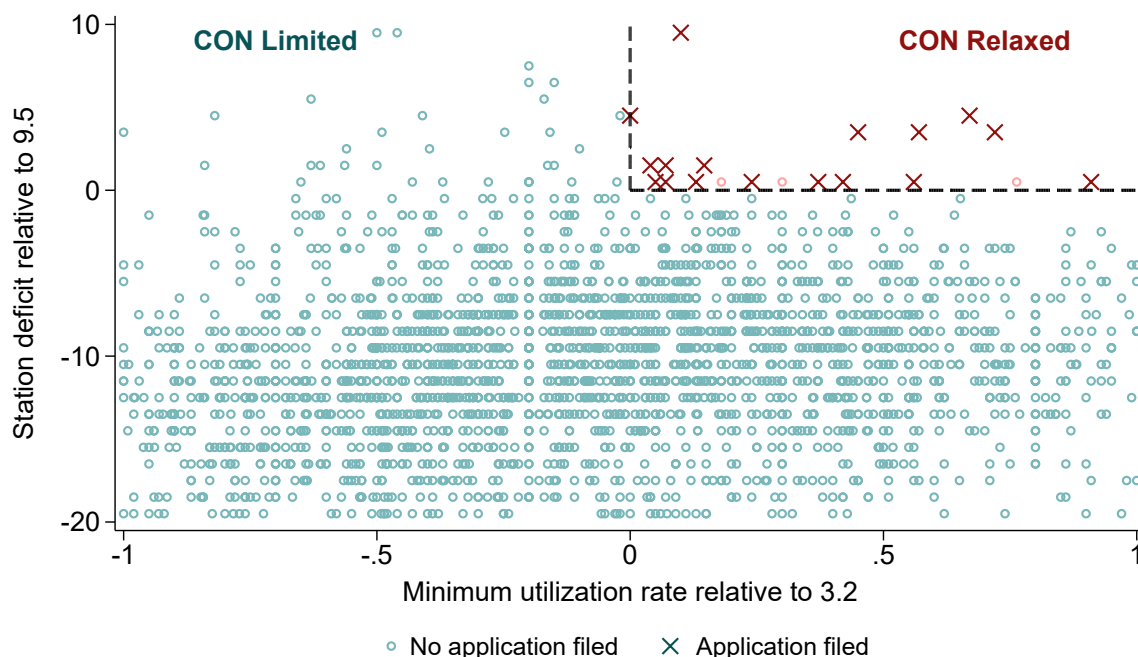


(a) Application-filing regression discontinuity

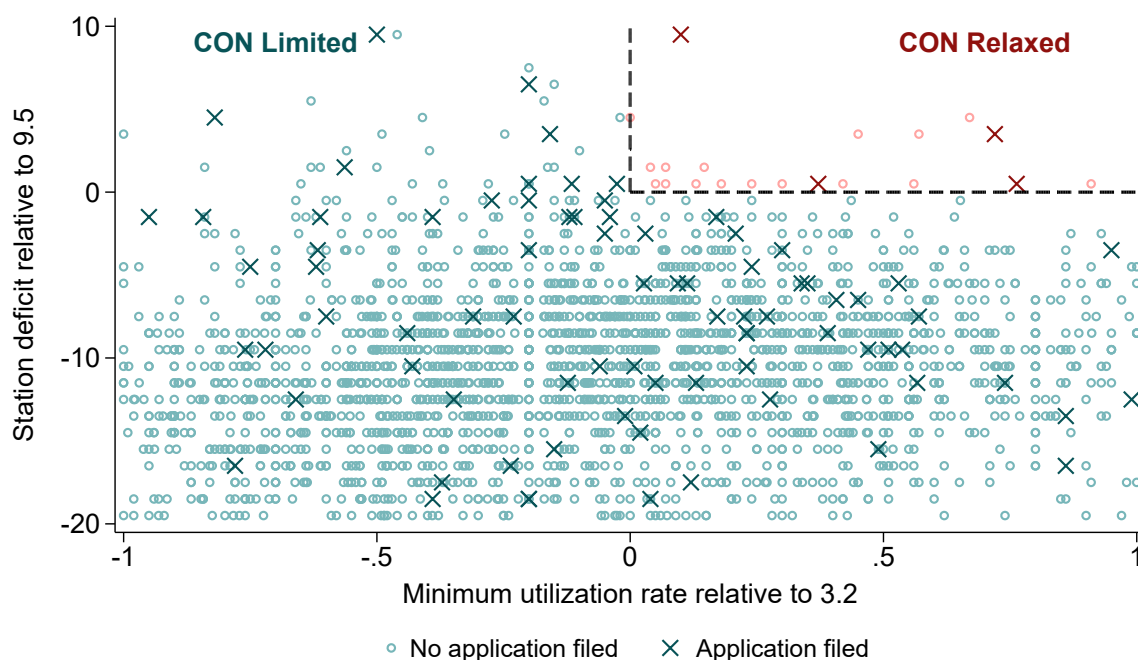


(b) Station deficit time series

Fig. 1. *Effect of regulatory thresholds in NC counties without incumbents.* This figure plots data for the sample of NC counties that did not have a dialysis center in 1997. Panel (a) plots the rate that potential entrants file applications as a function of county station deficits. Panel (b) plots trends in county station deficits and identifies application-filing dates and entry. See the discussion near page 12 and the regression discontinuity analysis in appendix C. Source: Author's analysis of the NC dialysis CON program's application data and dialysis reports.



(a) Applications to enter with all new stations



(b) Applications to enter with existing stations

Fig. 2. *Effect of regulatory thresholds in NC counties with incumbents.* This figure plots data for the sample of NC county-half-years with a center between 1997 and 2019. Both panels plot the joint distribution of station deficits and minimum utilization rates. Panel (a) identifies when applications to open all-new centers were filed. Panel (b) identifies when applications to open centers using some existing stations were filed. See the discussion near page 12. Source: Author's analysis of the NC dialysis CON program's application data and dialysis reports.

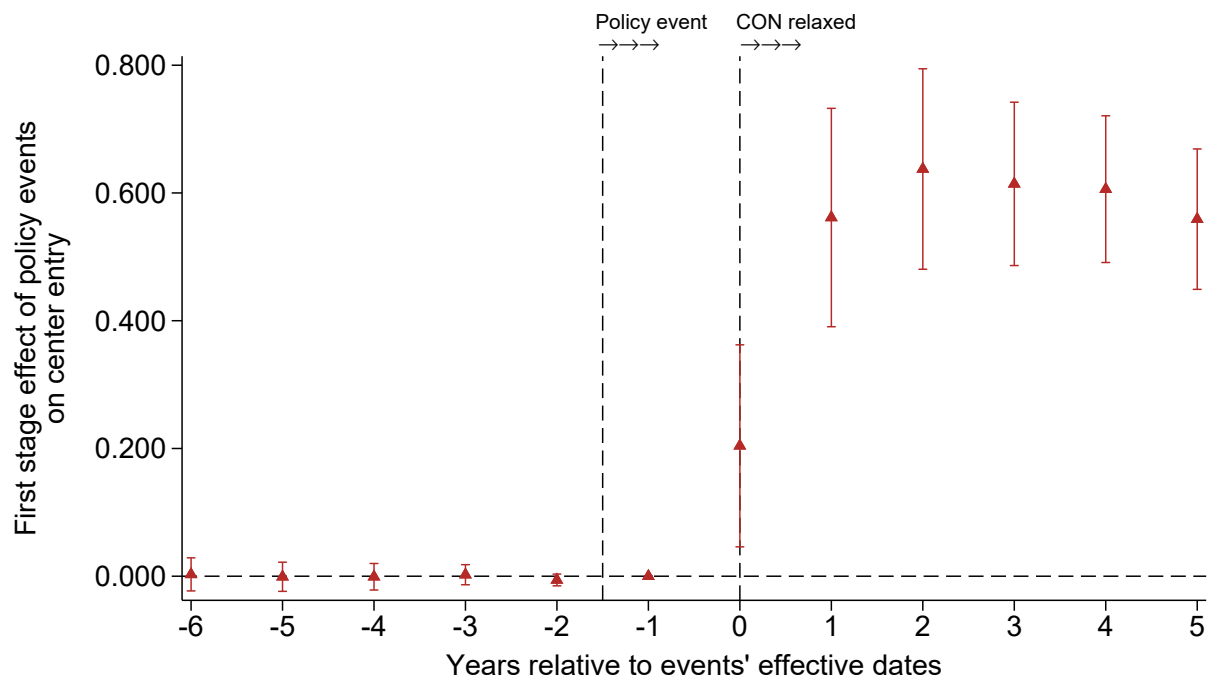


Fig. 3. *First stage effect of WA and NC dialysis CON events on entry.* This figure plots the effect of the WA and NC dialysis CON programs' events on the fraction of patients with a center in their county. The points are DID estimates of $(\delta_t : \tau \in \mathcal{W})$ from equation (5). The bars are 95% confidence intervals computed using county-level cluster-robust standard errors. See the discussion near page 16. Source: Author's analysis of the USRDS Database.

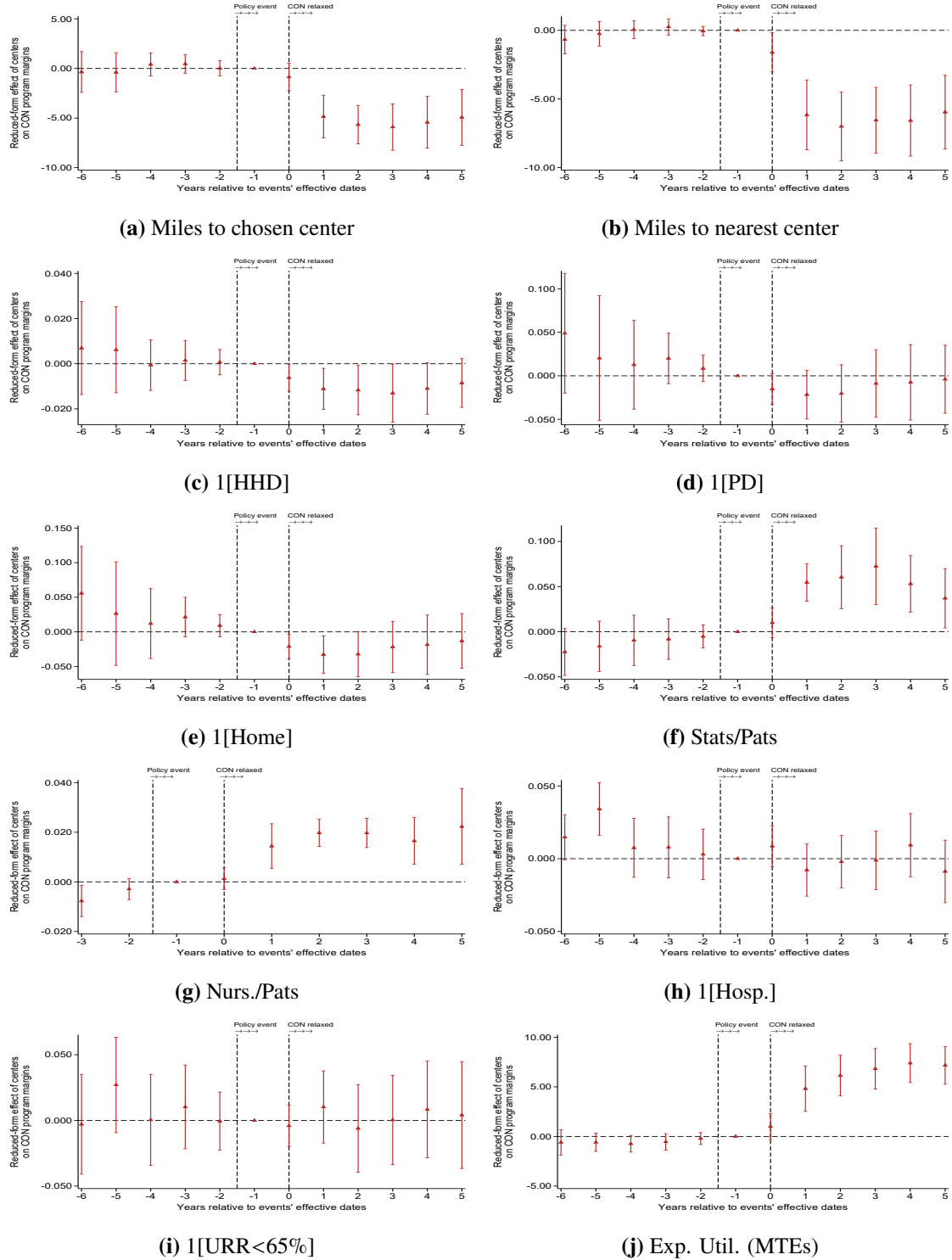
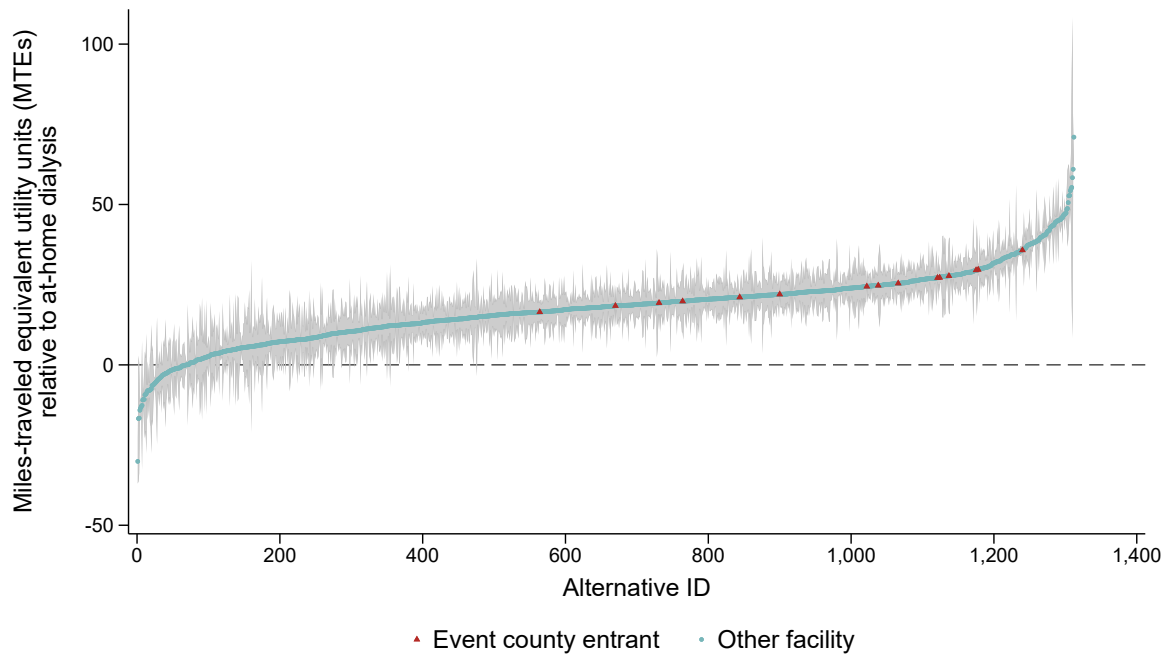


Fig. 4. DID estimates of WA and NC dialysis CON events on patient access, health, and welfare.

This figure reports the reduced-form results of the IV-DID analysis. Each panel plots estimates of equation (4) for various outcomes. Bars are 95% confidence intervals estimated with county-level cluster-robust standard errors. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). Comparison groups are nationwide or regional counties without centers on the events' effective dates. I use a shorter pre-event window for the nurses per patient outcome because staffing data are not available before 2004. See the discussions near pages 16 and 19. Source: author's analysis of the USRDS Database.

	(1)
TrDist	-0.109*** (0.002)
1[Stay]	5.875*** (0.036)
#[Cases]	1,368,224
#[Alts.]	1,312
<i>N</i>	27,881,747
#[Clusters]	447
#[Alts. per case]	
Avg.	20
Min.	2
Max.	93

(a) Travel and switching cost estimates



(b) Distribution of alternative-specific fixed effect estimates

Fig. 5. *Estimates of treatment choice model.* Panel (a) reports estimates of γ and λ in equation (6). County-level cluster-robust standard errors in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel (b) plots the distribution of alternative-specific fixed effects scaled by $-\hat{\gamma}$. Shaded areas are 95% confidence intervals. See the discussion near page 19. Source: author's analysis of the USRDS database.

ONLINE APPENDIX

Summary of sections in the online appendix

- Appendix [A](#) describes how I constructed key variables.
- Appendix [B](#) reports additional descriptive statistics.
- Appendix [C](#) describes the regression discontinuity analysis of the NC deficit threshold.
- Appendix [D](#) describes details about the 2007 WA policy change and NC threshold-crossings.
- Appendix [E](#) describes how IV-DID identifies CATE-ES in this context.
- Appendix [F](#) reports IV-DID estimates for additional outcomes
- Appendix [G](#) describes the results of the sensitivity analyses.

A Key Variables

As described in section 3, this study is based primarily on data from the U.S Renal Data System 2018 Database (USRDS Database) (USRDS 2018). The USRDS Database contains a number of standard analysis files (SAFs) produced by the USRDS using data from the Centers for Medicare and Medicaid Services (CMS), including Medicare claims, the chronic renal disease medical evidence reports, death notification reports, and dialysis center surveys. It also contains institutional Medicare claims data (USRDS IMCD). See USRDS (2018) for more information about the USRDS Database. Other vintages of the USRDS Database have been used in several studies, including Eliason et al. (2022) and Eliason et al. (2020). I also rely on data from the NC dialysis CON program, CMS's Healthcare Cost Report Information System (HCRIS), and several ancillary sources, including the National Bureau of Economic Research (NBER)'s ZCTA distances database, the NBER and Centers for Medicare and Medicaid Services (CMS)'s diagnosis-related group (DRG) to Major Diagnostic Category (MDC) crosswalks, the UDS Mapper ZIP-to-ZCTA crosswalk, and the U.S. Census Bureau's ZCTA-to-county relationship file.

In this section, I describe how I used these data to generate this study's key variables.

A.1 Patients' dialysis centers and modalities

The USRDS Database's RXHist SAF contains spell-level data, where each observation contains a patient ID, center ID, modality, start date, and end date. Each spell within a patient-center pair is mutually exclusive. I used these data to construct a (patient, month, center)-level dataset containing for each patient-month an observation for every center where they had a dialysis spell.

Each patient-month-center observation is associated with a treatment modality. I coarsened the modality indicator into 7 groups: (1) any dialysis, (2) in-center hemodialysis (HD), (3) continuous ambulatory peritoneal dialysis (CAPD), (4) continuous cycling peritoneal dialysis (CCPD), (5) any peritoneal dialysis (PD), (6) home hemodialysis (HHD), and (7) any home dialysis (PD or HHD).

In (patient, month)-level data, I assign each patient-month to that center with the earliest spell start date. (Patients are often associated with multiple centers in a given month when they switch centers during that month.) I allow each patient-month to be associated with multiple modalities when applicable.

A.2 Patients' payers

The USRDS Database's Payer SAF contains spell-level data, where each observation contains a patient ID, payer category, start date, and end date. Each patient's spells are mutually exclusive. I used these data to construct a (patient, month)-level dataset indicating each patient's payer category

in that month and whether they were dual-eligible. When a patient had more than one payer in a given month, I assigned their payer in that month to be whichever had the earliest start date.

The payer categories are (1) HMO, (2) Medicare primary (parts A and B), (3) Medicare primary (other), (4) Medicare secondary with employer-sponsored group health plan (EGHP), (5) Medicare secondary with no EGHP, and (6) 90-day waiting period. (The waiting period refers to the 3 month delay between the onset of ESRD and Medicare coverage.) I defined a patient-month to have had FFS Medicare as a primary payer if it is associated with Medicare primary (parts A and B).

A.3 Patients' residences

The USRDS Database's Residence SAF contains spell-level data, where each observation contains a patient ID, ZIP code, county FIPS code, state FIPS code, start date, and end date. Each patient's spells are mutually exclusive. I used these data to construct a (patient, month)-level dataset indicating each patient's residence in that month. The start date of each patient's first observed spell is always missing. I treat it as beginning at the start of the sample period. Likewise, the end date of each patient's last observed spell is always missing. I treat it as ending at the end of the sample period. When a patient had more than one residence in a given month, I assigned their residence in that month to be whichever had the earliest start date.

A.4 ZCTA-to-county crosswalk

The U.S. Census Bureau's 2010 ZCTA-to-county relationship file contains (ZCTA, county)-level data, where each observation contains the number of residents shared between the ZCTA and the county. I used these data to create a ZCTA-level crosswalk between ZCTAs and counties by linking each ZCTA to the county where the largest share of its population resided.

A.5 ZIP-to-ZCTA crosswalk

The UDS Mapper is an online data clearinghouse affiliated with the U.S. Department of Health & Human Services' Health Resources & Services' Administration. It publishes a ZIP-to-ZCTA crosswalk linking ZIP codes to ZCTAs. I downloaded the crosswalk in June 2020. It contained data for years 2010-2019. I used the 2010 subsample for years prior to 2010.

A.6 Dialysis centers' characteristics

The USRDS Database's Facility SAF contains (center, year)-level data, where each observation contains a variety of business records, including the center's ZIP code, state FIPS code, certification

date, number of stations, staffing, for-profit indicator, chain affiliation (if any), and a hospital-based indicator in that year. I combined these data with the RXHist SAF to construct a (center, month)-level dataset with an observation for each center-month in the RXHist SAF.

First, since the Facility SAF identifies each center-year's ZIP code but not its county, I used the ZIP-to-ZCTA crosswalk and the ZCTA-to-county crosswalk to link each center-year with a county. Second, when a center-year in the Facility SAF has a missing chain affiliation, I assume that it is an independent (non-chain) center.

Third, the Facility SAF contains staffing data beginning in 2004, including the number of full-time and part-time registered nurses (RNs), licensed practical nurses (LPNs), advanced practice nurses (APNs), patient care technicians (PCTs), social workers, and dieticians. Within each job category, I computed the total number of staff as the sum of the number of full-time staff plus 0.5 times the number of part-time staff. I also summed the number of RNs, LPNs, and APNs to compute the total number of nurses.

Fourth, when a variable has a missing value for a given center-year, I fill it in with non-missing values associated with that center in adjacent years. (E.g., if a center is observed to have had 10 stations in 2010 and missing stations in 2009, I assume that it also had 10 stations in 2009.) I leave staffing variables as missing prior to 2004.

Fifth, I used the RXHist SAF to compute the number of patients treated at each center-month. I combined this measure with the stations and staffing data derived from the Facility SAF to compute each center-month's stations-per-patient and staff-per-patient.

Finally, I used the RXHist SAF to determine each center's opening date. In particular, I treat each center's opening date as the earlier of (1) their earliest observed certification date in the Facility SAF and (2) their earliest observed treatment date in the RXHist SAF. For each (center, month), I compute the center's age in that month relative to this opening date.

A.7 Dialysis centers' reported costs

The HCRIS data contains (center, fiscal year)-level data, where each observation contains dialysis centers' reported costs in one of several cost categories in a given fiscal year. I define a dialysis center's reported fixed cost as the sum of the costs reported in lines 1-4, 6, and 11 of Form CMS-265-11 Worksheet A. These lines encompass:

1. Line 1: Capital related costs - buildings and fixtures
2. Line 2: Capital related costs - movable equipment
3. Line 3: Operation & maintenance of plant

4. Line 4: Housekeeping
5. Line 6: Machine capital-related or rental & maintenance
6. Line 11: administrative & general

I evenly distribute their reported fixed costs across each day in their fiscal year and then aggregate them up to the (center, month)-level.

A.8 Travel distances

The NBER ZCTA distances database contains (ZCTA1, ZCTA2)-level data, where each observation contains the distance between the ZCTAs' centroids. Only ZCTA pairs that are within 100 miles of each other are included. I used this database to measure distances between dialysis centers' locations, and between dialysis centers' locations and patients' residences. In both cases, I used the UDS Mapper's ZIP-to-ZCTA crosswalk to determine the centers' and patients' ZCTAs. If a ZCTA pair is not observed in the ZCTA distances database but each ZCTA in the pair is observed in the database, then I set the travel distance for that pair to be 100 miles. Otherwise, I treat the distance as missing.

A.9 Claims-based measures

The IMCD contains Medicare claim-level data, where each observation contains a patient ID, provider ID, Health Care Finance Administration (HCFA) source of the bill, type of bill, claim start date, claim end date, diagnosis related group (DRG), and total Medicare paid amount. The data include Medicare claims from 1977-2016, but some data are missing prior to 1989.

I used these data to construct several claims-based measures. Throughout, I define claims-based measures for only those patient-months associated with FFS Medicare as the primary payer. I treat them as missing otherwise because they may have been covered by another payer. (For instance, a patient-month associated with HMO coverage may not have an observed hospital stay either because no hospital stay occurred or because it was paid for by a non-Medicare plan.)

A.9.1 Hospitalizations

I defined a claim to be a hospital claim if its HCFA source is an inpatient hospital stay or if its HCFA source is an outpatient stay but its type of bill indicates that it was an inpatient hospital stay. I defined a patient-month to have a hospitalization if I observed a hospital claim associated with that patient in that month. Since 1989, each claim in the IMCD is associated with a diagnosis-related group (DRG). I categorized hospital stays using the DRG to Major Diagnostic Category (MDC)

crosswalk published by the NBER (1984-2013) and CMS (2014-2019) to link each claim's DRG with an MDC. There are 25 MDCs, including "circulatory system" and "infectious and parasitic diseases and disorders."

A.9.2 Medicare spending

Since 1989, each claim in the IMCD is associated with a total Medicare paid amount. I distributed each claim's total Medicare paid amount across the months associated with that claim in proportion to the number of that claim's days spent in that month. For instance, for a claim spanning March 28 through April 8, inclusive, I assigned one-third of the claim's paid amount to March and two-thirds to April. For each patient-month, I computed the total paid amount associated with that patient-month across all of their claims.

A.9.3 Dialysis sessions

Since 1989, some claims in the IMCD are associated with a number of dialysis sessions and indicators of treatment modality. I distributed each claim's sessions across the months associated with that claim in proportion to the number of that claim's days spent in that month. For instance, for a claim spanning March 28 through April 8, inclusive, I assigned one-third of the claim's sessions to March and two-thirds to April. For each patient-month, I computed the total number of sessions per modality associated with that patient-month across all of their claims. This process resulted in an improbable number of sessions for some patient-months (e.g., hundreds or thousands of sessions). When a patient-month is associated with more than 40 in-center HD or HHD sessions or more than 60 PD sessions, I treat these values as missing.

A.9.4 Urea reduction ratios

From 1998-2011, some claims in the IMCD are associated with a measure of their patients' urea reduction ratio (URR). The measure indicates whether the URR was <60%, 60-65%, 65-70%, 70-75%, or >75%. From 2012-2016, these indicators are in the USRDS Claims Clinical SAF. I append the data from these sources together to generate (patient, month)-level data spanning 1998-2016 and indicating whether each patient had a URR within any of the given ranges in that month.

A.10 Patients' demographics and other characteristics

The USRDS Database's Patient SAF contains records of each patient's sex, race, ethnicity, primary cause of ESRD, and date of birth. I generate an indicator for each sex, race, ethnicity, and primary

cause of ESRD category. I also measure each patient's age in each month using their date of birth. I set age to missing for any patients ever associated with an age below 0 or greater than 105.

The USRDS Database's Death SAF contains records of each patient's date of death, if any. I define a patient to have died in a given month if their date of death falls in that month.

The USRDS Database's Medical Evidence SAF contains records of each patient's employment status, height, weight, and BMI at ESRD onset. I generate patient-level indicators for any employment (full-time or part-time), no employment (unemployed, homemaker, retired, medical leave-of-absence, and student), and other. I also generate patient-level measures of each patient's height, weight, and BMI. Some heights and weights are implausible for some patients (e.g., 0 kilograms or more than 1,000 kilograms). I set these to missing.

Finally, the USRDS Database's Transplant SAF contains records of each patient that ever received a kidney transplant.

A.11 North Carolina station deficits, minimum utilization rates, and application data

The NC dialysis CON program publishes facility-level and county-level data in semiannual dialysis reports (SDRs). The facility-level data include each center's patient volume, number of dialysis stations, and utilization rate. The county-level data include each county's current, lagged, and forecasted resident in-center dialysis patient population, number of dialysis stations, and station deficit. I collected SDRs spanning 1997-2019 and manually entered their facility-level and county-level data to produce a (county, half-year)-level dataset containing each county's station deficit and minimum utilization rate.

I also rely on contemporaneous records of applications for certificates-of-need filed with the NC dialysis CON program by existing centers and potential entrants. The data include the date that each application was filed, the applicant's name, the result of the review process (e.g., if the certificate-of-need was granted or denied), the project's location (usually this is the name of a dialysis center and a county), the project's projected cost, and the purpose of the project (e.g., to open a new center or add stations to an existing center). I manually reviewed all of the project descriptions and identified applications to open a center using all-new stations or using stations that had previously been approved at another location. Since applications may be filed in semiannual application cycles that begin after the publication of an SDR, I associate each application with the most recently published prior SDR.

B Additional descriptive statistics

In section [3](#), I reported descriptive statistics for the sample of county-months, patient-months, and center-months that I constructed from the USRDS Database and ancillary data sources. I reported the descriptive statistics for years 2004-2016 to align them with the cross-sectional analysis in section [4](#). In this section, I report descriptive statistics for the entire 1980-2016 sample period. The results are in tables [A3-A7](#).

C Station deficit threshold regression discontinuity

In section 5, I showed graphically that potential entrants file applications to open new dialysis centers in North Carolina counties without incumbents when their station deficits exceed 9.5. In this section, I substantiate this finding with a regression discontinuity analysis.

Define the index c for counties and t for half-years. Let d_{ct} be the station deficit of county c in half-year t , and define $\tilde{d}_{ct} := d_{ct} - 9.5$. I construct a (county, half-year)-level panel dataset, consisting of the 24 North Carolina counties that did not have an operational or approved center as of the March 1997 SDR. I estimate the model:

$$Y_{ct} = \beta 1[\tilde{d}_{ct} \geq 0] + \delta_1 \tilde{d}_{ct} + \delta_2 \tilde{d}_{ct} 1[\tilde{d}_{ct} \geq 0] + \varepsilon_{ct} \quad (\text{C.1})$$

where Y_{ct} indicates that an application to open a new center in county c was filed in half-year t . The parameter of interest is β , which measures the discontinuous change in Y_{ct} associated with exceeding the station deficit threshold. I estimate β using mean-squared error optimal bandwidths and robust confidence intervals, following [Calonico et al. \(2014a\)](#) and [Calonico et al. \(2014b\)](#).

Table A8 presents the results. It shows that when North Carolina counties without incumbents exceed the station deficit threshold, application-filing rates increase by 55 percentage points.

D Additional details about the 2007 WA reform and NC threshold-crossings

The NC dialysis CON program permits entry in counties without incumbents when their station deficits exceed 9.5. I obtained SDRs between 1997 and 2019. According to these records, 24 counties did not have an operating or previously-approved center in March 1997. Of these counties, Alexander, Davie, Gates, Greene, Haywood, Jones, Perquimans, Polk, Stokes, Swain, Warren, Washington, and Yadkin counties experienced a threshold-crossing between 1997 and 2019. I exclude Perquimans County from the IV-DID analysis because it crossed the station deficit threshold in January 2014 and the sample period ends shortly thereafter.

In Washington, a 2007 reform relaxed the WA dialysis CON program's entry limits in counties without incumbents in 2006. The target WA counties were Adams, Columbia, Douglas, Ferry, Garfield, Jefferson, Kittitas, Klickitat, Lincoln, Okanogan, Pacific, Pend Oreille, San Juan, Skamania, Stevens, and Wahakium counties ([WA State Register 2006](#)). However, according to the USRDS Database, Okanogan County had a center in 2006. Furthermore, according to records from the WA dialysis CON program, someone filed an application to open a new center in Kittitas County shortly before 2007. Although the application was reviewed after 2007, it was reviewed using pre-2007 rules. Therefore, I exclude Okanogan and Kittitas counties from the IV-DID analysis.

E Instrumental variables difference-in-differences

I leverage variation generated by the 2007 WA policy change and the NC threshold-crossings using instrumental variables difference-in-differences (IV-DID). This approach identifies the complier average treatment effect at the event site (CATE-ES) under the following assumptions.

Consider a setting with two periods $t \in \{1, 2\}$ and a mass of units (e.g., counties). Let there be some event (i.e., the 2007 WA policy change or a NC threshold-crossing) that is related to some treatment (i.e., having a dialysis center) and some outcome (e.g., health status). Let $Z_i \in \{0, 1\}$ indicate whether some unit i is exposed to the event. Let $D_i(Z) \in \{0, 1\}$ indicate unit i 's potential treatment status in period 2. Since the event counties and comparison counties did not have dialysis centers prior to the events, we may assume that no units are exposed to the treatment in period 1 for simplicity. Finally, let $Y_{it}(D, Z)$ measure unit i 's potential outcome in period t . Assume that $Y_{it}(D, Z) = Y_{it}(D)$ for all (D, Z) ("exclusion restriction").

Let $C := \{i : D_i(1) - D_i(0) = 1\}$ be the set of compliers. These units would only experience the treatment if they experience the event. Let $\mathcal{D} := \{i : D_i(1) - D_i(0) = -1\}$ be the set of defiers. These units would only experience the treatment if they do not experience the event. Assume that $\mathcal{D} = \emptyset$ ("monotonicity"). Assume that $C \neq \emptyset$ ("relevance").

Let $(Z_i, D_i, Y_{i1}, Y_{i2})$ be unit i 's realized exposure to the event, treatment status, and outcomes. Assume that $\mathbb{E}[D_i(0)|Z_i = 1] = \mathbb{E}[D_i(0)|Z_i = 0]$ ("first stage mean independence"). This states that the average counterfactual treatment status in the event counties is equal to the average realized treatment status in the comparison counties. Assume that $\mathbb{E}[Y_{i1}(D_i(1))|Z_i = 1] = \mathbb{E}[Y_{i1}(D_i(0))|Z_i = 1]$ ("reduced form no anticipation"). This states that the average realized period 1 outcome in the event counties is equal to the average counterfactual period 1 outcome in the event counties. Finally, assume that $\mathbb{E}[Y_{i2}(D_i(0)) - Y_{i1}(D_i(0))|Z_i = 1] = \mathbb{E}[Y_{i2}(D_i(0)) - Y_{i1}(D_i(0))|Z_i = 0]$ ("reduced form parallel trends"). This states that the average counterfactual outcome trend in the event counties is equal to the average realized outcome trend in the comparison counties.

The reduced form effect is $\mathbb{E}[Y_{i2}(D_i(1)) - Y_{i2}(D_i(0))|Z_i = 1]$. It is identified by:

$$\begin{aligned}
 & \mathbb{E}[Y_{i2} - Y_{i1}|Z_i = 1] - \mathbb{E}[Y_{i2} - Y_{i1}|Z_i = 0] & (E.2) \\
 &= \mathbb{E}[Y_{i2}(D_i(1))|Z_i = 1] - \mathbb{E}[Y_{i1}(D_i(1))|Z_i = 1] - \mathbb{E}[Y_{i2}(D_i(0)) - Y_{i1}(D_i(0))|Z_i = 0] \\
 &= \mathbb{E}[Y_{i2}(D_i(1))|Z_i = 1] - \mathbb{E}[Y_{i1}(D_i(0))|Z_i = 1] - \mathbb{E}[Y_{i2}(D_i(0)) - Y_{i1}(D_i(0))|Z_i = 1] \\
 &\quad \text{by reduced form no anticipation and parallel trends} \\
 &= \mathbb{E}[Y_{i2}(D_i(1)) - Y_{i2}(D_i(0))|Z_i = 1]
 \end{aligned}$$

Likewise, the first stage effect is $\mathbb{E}[D_i(1) - D_i(0)|Z_i = 1]$. It is identified by:

$$\begin{aligned}
 & \mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0] \\
 &= \mathbb{E}[D_i(1)|Z_i = 1] - \mathbb{E}[D_i(0)|Z_i = 0] \\
 &= \mathbb{E}[D_i(1)|Z_i = 1] - \mathbb{E}[D_i(0)|Z_i = 1] \\
 &\quad \text{by first stage mean independence} \\
 &= \mathbb{E}[D_i(1) - D_i(0)|Z_i = 1]
 \end{aligned} \tag{E.3}$$

Finally, the parameter of interest is CATE-ES $:= \mathbb{E}[Y_{i2}(1) - Y_{i2}(0)|i \in C, Z_i = 1]$. It is identified under the foregoing assumption by the ratio of the reduced form effect (E.2) and the first stage effect (E.3), as follows:

$$\begin{aligned}
 & \frac{\mathbb{E}[Y_{i2} - Y_{i1}|Z_i = 1] - \mathbb{E}[Y_{i2} - Y_{i1}|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]} \\
 &= \frac{\mathbb{E}[Y_{i2}(D_i(1)) - Y_{i2}(D_i(0))|Z_i = 1]}{\mathbb{E}[D_i(1) - D_i(0)|Z_i = 1]} \\
 &= \frac{\mathbb{E}[Y_{i2}(1) - Y_{i2}(0)|i \in C, Z_i = 1]\mathbb{P}(i \in C|Z_i = 1)}{\mathbb{P}(i \in C|Z_i = 1)} \\
 &\quad \text{by monotonicity and relevance} \\
 &= \mathbb{E}[Y_{i2}(1) - Y_{i2}(0)|i \in C, Z_i = 1]
 \end{aligned} \tag{E.4}$$

F Additional IV-DID outcomes

In section 6.4, I reported IV-DID results for several access, treatment choice, and health outcomes. Table A9 and figure A3 report results for several additional outcomes.

G Sensitivity analysis

In section 6.5 and 6.6, I described the results of several sensitivity analyses. In this section, I report the results of the sensitivity analyses in full.

G.1 IV-DID sensitivity to control variables

Table A10 reproduces the IV-DID analysis, excluding patient-level control variables.

G.2 IV-DID sensitivity to comparison counties

Table A11 reproduces the IV-DID analysis, replacing the national comparison counties with regional comparison counties.

G.3 IV-DID sensitivity to pooling WA and NC

Table A12 reproduces the IV-DID analysis for the 2007 WA reform only. Table A13 reproduces the IV-DID analysis for the NC threshold-crossings only.

G.4 Choice model sensitivity to linear travel cost

Figure A4 reproduces the choice model estimates using a quadratic measure of patients' distances to their alternatives. That is, I estimate

$$u_{ijt} = \gamma_1 \text{TrDist}_{ijt} + \gamma_2 \text{TrDist}_{ijt}^2 + \lambda_j \text{Stay}_{ijt} + \delta_{jt} + \varepsilon_{ijt} \quad (\text{G.5})$$

As in section 6.6, I assume that $\lambda_j = \lambda$ and $\delta_{jt} = \delta_j$ for all j, t in order to reduce the computational cost.

County (1)	Stations (2)	Policy Regime (3)	Projected Patients Needed For Given Num. of Stations (4)	Resident Patient Populations from the Target WA Counties (pre) and the Neighboring Target WA Counties (post)					
				2007 (5)	2008 (6)	2009 (7)	2010 (7)	2012 (8)	2015 (9)
1. Stevens	10	Pre	48	12	15	20	25	29	38
		Post	32	31	36	44	41	51	70
2. Adams	5	Pre	24	15	17	15	17	17	19
		Post	16	20	22	21	22	20	28
3. Pacific	10	Pre	48	22	23	24	24	17	24
		Post	32	28	26	28	28	22	28
4. Jefferson	6	Pre	29	13	<11	<11	11	16	18
		Post	19	14	<11	11	13	16	20
5. Douglas	8	Pre	38	34	35	31	30	33	31
		Post	26	34	35	31	30	33	31

Tab. A1. *Characteristics of entrants in the target WA counties.* This table presents characteristics of the centers that opened in the target WA counties after the 2007 reform and the counties' resident patient populations. Prior to the reform, station deficits were computed using each county's resident in-center dialysis patient population and an assumption that each station could serve 4.8 patients. After the reform, station deficits were computed using each county's resident in-center dialysis patient population, each neighboring target WA county's resident in-center dialysis patient population while the neighboring county does not have a center, and an assumption that each station could serve 3.2 patients. Squares indicate years when the new centers were operational. Statistics are omitted for cells that consist of fewer than 11 individuals. See the discussion near page 16. Source: Author's analysis of the USRDS Database and the Medicare Dialysis Facility Compare files.

	Reported Fixed Costs
#[HD Patients]	0.011593*** (0.000417)
#[HD Patients] ²	-0.000013*** (0.000003)
#[Home Patients]	0.008636*** (0.000321)
#[Home Patients] ²	-0.000011*** (0.000002)
#[Stations]	0.026473*** (0.001907)
#[Stations]×#[HD Patients]	-0.000123*** (0.000024)
%[County rural pop.]	-0.001472*** (0.000166)
Age FE	Y
Chain-by-trend FE	Y
State FE	Y
<i>N</i>	400,395
#[Clusters]	6,452
Baseline <i>Y</i>	91,624
<i>R</i> ²	0.757

Tab. A2. *Estimates of reported fixed cost prediction model.* This table presents estimates of a linear regression model relating log reported fixed costs to dialysis center characteristics. Baseline *Y* is the sample average value of the outcome. See the discussion near page 20. Source: Author's analysis of the USRDS Database and HCRIS data.

	N (1)	Mean (2)	SD (3)	5%ile (4)	Median (5)	95%ile (6)
Capacity and resident patient population						
1[Has Center] ¹	1,395,012	0.42	0.49	0.00	0.00	1.00
#[Centers] ¹	1,395,012	1.40	4.98	0.00	0.00	6.00
#[Stations] ¹	1,395,012	21.44	87.09	0.00	0.00	90.00
#[Patients] ¹	1,395,012	76.82	346.07	0.00	15.00	286.00
Facility ownership						
#[Chain-owned] ¹	1,395,012	0.36	0.72	0.00	0.00	2.00
#[Independent] ¹	1,395,012	0.70	2.74	0.00	0.00	3.00
#[Owners] ¹	1,395,012	1.06	3.06	0.00	0.00	4.00
Sample composition						
#[County-months]	1,395,012	1,395,012	1,395,012	1,395,012	1,395,012	1,395,012
#[Counties]	3,144	3,144	3,144	3,144	3,144	3,144

Tab. A3. *Counties' dialysis patient and center characteristics during 1980-2016.* See the discussion near page 48. Source: author's analysis of the USRDS Database.

¹ Estimated from a (county, month)-level sample.

	N (1)	Mean (2)	SD (3)	5%ile (4)	Median (5)	95%ile (6)
Age, race, ethnicity, and sex						
Age ¹	108,891,800	60.10	15.63	32.00	62.00	83.00
1[White] ²	2,852,343	0.66	0.47	0.00	1.00	1.00
1[Black] ²	2,852,343	0.29	0.45	0.00	0.00	1.00
1[AIAN] ²	2,852,343	0.01	0.10	0.00	0.00	0.00
1[Asian] ²	2,852,343	0.03	0.17	0.00	0.00	0.00
1[NHPI] ²	2,852,343	0.01	0.09	0.00	0.00	0.00
1[Hispanic] ²	2,852,343	0.11	0.31	0.00	0.00	1.00
1[Not hispanic] ²	2,852,343	0.75	0.43	0.00	1.00	1.00
1[Female] ²	2,852,343	0.45	0.50	0.00	0.00	1.00
1[Male] ²	2,852,343	0.55	0.50	0.00	1.00	1.00
Employment status at ESRD onset						
1[Some empl.] ²	2,852,343	0.08	0.27	0.00	0.00	1.00
1[No empl.] ²	2,852,343	0.67	0.47	0.00	1.00	1.00
Height and weight at ESRD onset						
Height (cm) ²	2,162,934	168.00	11.62	150.00	168.00	185.42
Weight (kg) ²	2,171,303	79.75	24.26	47.63	76.00	124.00
BMI ²	2,127,998	28.35	7.81	18.50	26.87	43.21
Primary cause of ESRD						
1[Diabetes] ²	2,852,343	0.42	0.49	0.00	0.00	1.00
1[Hypertension] ²	2,852,343	0.28	0.45	0.00	0.00	1.00
1[Glomer.] ²	2,852,343	0.11	0.31	0.00	0.00	1.00
1[Cyst. Kidney] ²	2,852,343	0.02	0.15	0.00	0.00	0.00
Sample composition						
#[Patient-months]	108,988,574	108,988,574	108,988,574	108,988,574	108,988,574	108,988,574
#[Patients]	2,852,343	2,852,343	2,852,343	2,852,343	2,852,343	2,852,343

Tab. A4. *Patients' pre-dialysis characteristics during 1980-2016.* The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussion near page 48. Source: author's analysis of the USRDS Database.

¹ Estimated from a (patient, month)-level sample.

² Estimated from a patient-level sample.

	N (1)	Mean (2)	SD (3)	5%ile (4)	Median (5)	95%ile (6)
Miles to nearest alternatives						
Any alternative ¹	107,808,188	4.55	7.82	0.00	1.96	18.94
≥50% Stats/Pats ¹	107,808,188	9.15	13.50	0.00	4.36	33.97
Alternatives within 100 miles						
#[Centers] ¹	107,786,177	135.78	116.74	14.00	100.00	409.00
#[Chains] ¹	107,786,177	4.53	2.14	1.00	5.00	8.00
#[Independent] ¹	107,786,177	53.97	59.96	4.00	31.00	208.00
#[Owners] ¹	107,786,177	58.50	61.03	6.00	35.00	216.00
Chosen center characteristics						
Distance (mi) ¹	106,537,421	11.82	20.18	0.00	5.40	52.05
#[Stats/Pats] ¹	107,718,973	0.25	0.48	0.11	0.21	0.44
#[Nurses/Pats] ^{1,2}	62,248,996	0.07	0.11	0.03	0.06	0.15
#[PCTs/Pats] ^{1,2}	62,248,996	0.08	0.10	0.01	0.08	0.13
#[Dietic./Pats] ^{1,2}	62,248,996	0.01	0.01	0.00	0.01	0.02
1[Not switched] ¹	106,185,352	0.98	0.15	1.00	1.00	1.00
1[Switched] ¹	106,185,352	0.02	0.15	0.00	0.00	0.00
Treatment modalities						
1[In-center HD] ¹	108,988,574	0.89	0.32	0.00	1.00	1.00
1[PD] ¹	108,988,574	0.11	0.31	0.00	0.00	1.00
1[HHD] ¹	108,988,574	0.01	0.12	0.00	0.00	0.00
1[Ever TX] ³	2,852,343	0.14	0.34	0.00	0.00	1.00
Sample composition						
#[Patient-months]	108,988,574	108,988,574	108,988,574	108,988,574	108,988,574	108,988,574
#[Patients]	2,852,343	2,852,343	2,852,343	2,852,343	2,852,343	2,852,343

Tab. A5. *Patients' treatment choices during 1980-2016.* The sum of shares may exceed 1 due to rounding. Shares of “other” and “unknown” categories are omitted. See the discussion near page 48. Source: author's analysis of the USRDS Database.

¹ Estimated from a (patient, month)-level sample.

² Staffing data is only available in 2004-2016.

³ Estimated from a patient-level sample.

	N (1)	Mean (2)	SD (3)	5%ile (4)	Median (5)	95%ile (6)
Health insurance enrollment and spending						
1[MCare FFS Prim.] ¹	108,988,574	0.73	0.44	0.00	1.00	1.00
1[HMO] ¹	108,988,574	0.09	0.28	0.00	0.00	1.00
1[Any Medicare] ¹	108,988,574	0.81	0.39	0.00	1.00	1.00
1[Dual Eligible] ¹	108,988,455	0.33	0.47	0.00	0.00	1.00
Medicare spending ^{1,2,3}	73,238,152	4,590.17	6,981.34	69.00	2,471.00	16,737.09
Dialysis sessions						
#[In-center HD] ^{1,2,3}	73,201,631	10.46	4.90	0.00	13.00	14.00
#[HHD] ^{1,2,3}	73,229,958	0.15	1.62	0.00	0.00	0.00
#[PD] ^{1,2,3}	73,205,039	1.73	6.71	0.00	0.00	13.29
Urea reduction ratio (URR)						
1[URR <65%] ^{1,2,4}	50,141,537	0.09	0.29	0.00	0.00	1.00
1[URR 65-75%] ^{1,2,4}	50,141,537	0.46	0.50	0.00	0.00	1.00
1[URR >75%] ^{1,2,4}	50,141,537	0.44	0.50	0.00	0.00	1.00
Hospitalizations and mortality						
1[Any] ^{1,2}	73,238,152	0.17	0.38	0.00	0.00	1.00
1[Circulatory] ^{1,2,3}	73,238,152	0.07	0.25	0.00	0.00	1.00
1[Kidney and UT] ^{1,2,3}	73,238,152	0.02	0.15	0.00	0.00	0.00
1[Infections] ^{1,2,3}	73,238,152	0.01	0.12	0.00	0.00	0.00
1[Died] ¹	108,988,574	0.02	0.13	0.00	0.00	0.00
Sample composition						
#[Patient-months]	108,988,574	108,988,574	108,988,574	108,988,574	108,988,574	108,988,574
#[Patients]	2,852,343	2,852,343	2,852,343	2,852,343	2,852,343	2,852,343

Tab. A6. Patients' health outcomes and spending during 1980-2016. The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussion near page 48. Source: author's analysis of the USRDS Database.

¹ Estimated from a (patient, month)-level sample.

² Sub-sample associated with Medicare FFS as a primary payer.

³ Claims-based measures available in 1989-2016.

⁴ URR data available in 1998-2016.

	N (1)	Mean (2)	SD (3)	5%ile (4)	Median (5)	95%ile (6)
Age and chain affiliation						
Age ¹	1,841,559	14.38	10.96	0.92	11.92	36.25
1[Davita] ²	150,249	0.17	0.38	0.00	0.00	1.00
1[Fresenius] ²	150,249	0.21	0.41	0.00	0.00	1.00
1[Small chain] ²	150,249	0.14	0.35	0.00	0.00	1.00
1[Independent] ²	150,249	0.47	0.50	0.00	0.00	1.00
Profit status and hospital affiliation						
1[For-profit] ²	147,471	0.72	0.45	0.00	1.00	1.00
1[Not for-profit] ²	147,471	0.28	0.45	0.00	0.00	1.00
1[Freestanding] ²	148,097	0.78	0.42	0.00	1.00	1.00
1[Hosp. based] ²	148,097	0.22	0.42	0.00	0.00	1.00
Monthly patient volume						
#[Patients] ¹	1,841,559	60.47	52.90	1.00	49.00	160.00
#[In-center HD] ¹	1,841,559	53.55	46.31	1.00	45.00	140.00
#[HHD] ¹	1,841,559	0.84	5.12	0.00	0.00	4.00
#[PD] ¹	1,841,559	6.26	13.67	0.00	0.00	33.00
Capacity and monthly congestion						
#[Stations] ²	150,079	15.77	8.88	3.00	15.00	31.00
#[Stats/Pats] ¹	1,709,641	1.05	3.40	0.10	0.26	5.75
#[Nurses/Pats] ^{1,3}	894,724	0.16	0.74	0.03	0.07	0.31
#[PCTs/Pats] ^{1,3}	894,724	0.13	0.70	0.00	0.08	0.17
#[Dietic./Pats] ^{1,3}	894,724	0.03	0.09	0.00	0.01	0.06
Miles to nearest center						
To Co-owned ^{1,4}	922,646	14.65	18.97	0.00	7.80	50.24
To Competing ¹	1,709,105	8.99	14.85	0.00	2.64	38.32
Monthly reported costs (\$K)						
Total ^{1,5}	403,710	224,791.42	136,542.97	71,761.43	194,322.46	479,484.79
Capital ^{1,5}	403,710	36,959.64	20,505.16	10,862.15	33,797.77	72,003.74
Staff ^{1,5}	403,710	69,513.64	51,029.03	14,931.15	57,384.65	166,276.60
Supply ^{1,5}	403,710	23,927.80	21,268.38	4,867.47	17,766.48	64,340.49
Admin ^{1,5}	403,710	54,695.15	32,518.03	16,463.63	48,117.95	114,823.64
Other ^{1,5}	403,710	39,695.19	28,404.05	7,681.97	33,432.20	92,461.32
Fixed ^{1,5}	403,710	91,654.80	49,187.05	30,590.04	82,574.49	179,441.53
Average ^{1,5}	403,710	3,375.40	4,222.92	2,268.45	2,949.91	4,837.02
Sample composition						
#[Centers]	12,425	12,425	12,425	12,425	12,425	12,425
#[Center-years]	169,327	169,327	169,327	169,327	169,327	169,327
#[Center-months]	1,841,559	1,841,559	1,841,559	1,841,559	1,841,559	1,841,559

Tab. A7. *Centers' characteristics during 1980-2016.* The sum of shares may exceed 1 due to rounding. Shares of "other" and "unknown" categories are omitted. See the discussion near page 48. Source: author's analysis of the USRDS Database and HCRIS data.

¹ Estimated from a (center, month)-level sample.

² Estimated from a (center, year)-level sample.

³ Staffing data only available in 2004-2016.

⁴ Only defined for chain-owned centers.

⁵ Only available during 2011-2016.

	1[Application filed] (1)
RD	0.552** (0.219)
Bandwidth-L	2.0
Bandwidth-R	4.5
N	618
McCrary test (t)	-0.41
McCrary test (p)	0.68

Tab. A8. *Regression discontinuity estimates of the effect of NC threshold-crossings.* This table presents estimates of β from equation C.1. It corresponds to the regression discontinuity depicted in figure 1(a). See the discussion near page 49. Source: Author's analysis of the NC dialysis CON program SDRs and application data.

	HD Sess. for HD Pats. (1)	Medicare spending (2)	1[Died] (3)
1[Has Center]	-0.176 (0.226) [0.778]	-146.797 (181.688) [0.808]	0.002 (0.002) [1.164]
ZCTA FE	Y	Y	Y
Month FE	Y	Y	Y
Pat. Chars.	Y	Y	Y
Comp. group	USA	USA	USA
Baseline Y	12.114	4,004.713	0.018
Reduced form	-0.092	-77.248	0.001
First stage	0.523***	0.526***	0.533***
F-statistic	72.164	79.717	87.538
VtF 95% CV	1.910	1.915	1.921
#[Clusters]	1,741	1,743	1,744
#[Treated]	57,687	66,131	80,271
#[Untreated]	4,078,323	4,849,749	5,989,062
N	46,671,276	54,686,612	67,385,106

Tab. A9. *IV-DID estimates of marginal centers' effects on additional outcomes.* This table reports the additional results of the IV-DID analysis. The top panel reports estimates of β in equation (3a). County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). Comparison groups are nationwide counties without centers on the events' effective dates. #[Treated] and #[Untreated] refer to unique patient-months and N refers to all observations in the stacked dataset. See the discussions near pages 16 and 19. Source: author's analysis of the USRDS Database and HCRIS data.

	Mi. to Chosen Center (1)	Mi. to Nearest Center (2)	1[HHD] (3)	1[PD] (4)	1[Home] (5)
1[Has Center]	-8.572*** (1.599) [5.363]	-10.469*** (1.865) [5.612]	-0.022** (0.011) [2.030]	-0.048 (0.037) [1.313]	-0.071* (0.036) [1.939]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	N	N	N	N	N
Comp. group	USA	USA	USA	USA	USA
Baseline Y	25.755	18.244	0.015	0.162	0.176
Reduced form	-4.635***	-5.576***	-0.012**	-0.026	-0.038*
First stage	0.541***	0.533***	0.533***	0.533***	0.533***
F-statistic	94.947	87.233	87.233	87.233	87.233
VtF 95% CV	1.918	1.916	1.919	1.922	1.922
#[Clusters]	1,743	1,743	1,744	1,744	1,744
#[Treated]	78,797	80,271	80,271	80,271	80,271
#[Untreated]	5,867,165	5,988,933	5,989,062	5,989,062	5,989,062
N	65,847,120	67,383,307	67,385,108	67,385,108	67,385,108
	Stats/Pats At Chosen Center (6)	Nurs./Pats At Chosen Center (7)	1[Hosp.] (8)	1[URR <65%] (9)	Expected Utility (MTEs) (10)
1[Has Center]	0.105*** (0.021) [4.968]	0.037*** (0.006) [6.257]	-0.024* (0.013) [1.920]	-0.007 (0.027) [0.274]	10.515*** (0.884) [11.897]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	N	N	N	N	N
Comp. group	USA	USA	USA	USA	Reg.
Baseline Y	0.227	0.060	0.182	0.101	54.122
Reduced form	0.057***	0.019***	-0.013**	-0.003	6.008***
First stage	0.541***	0.500***	0.526***	0.446***	0.571***
F-statistic	94.948	43.530	79.544	30.706	83.364
VtF 95% CV	1.926	1.905	1.915	1.850	1.909
#[Clusters]	1,744	1,420	1,743	1,510	444
#[Treated]	78,797	31,496	66,131	36,536	78,797
#[Untreated]	5,867,331	2,874,369	4,849,749	2,883,666	1,276,452
N	65,849,229	33,349,406	54,686,614	34,143,755	8,453,182

Tab. A10. Sensitivity analysis of IV-DID estimates of marginal centers' effects on patient access, health, and welfare. This table reports a sensitivity analysis of the main results from the IV-DID analysis. Unlike the main results, these estimates are computed without patient characteristics. The top panel reports estimates of β in equation (3a). County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Comparison groups are nationwide or regional counties without centers on the events' effective dates. #[Treated] and #[Untreated] refer to unique patient-months and N refers to all observations in the stacked dataset. See the discussion near page 54. Source: author's analysis of the USRDS Database.

	Mi. to Chosen Center (1)	Mi. to Nearest Center (2)	1[HHD] (3)	1[PD] (4)	1[Home] (5)
1[Has Center]	-9.428*** (1.465) [6.437]	-10.760*** (1.586) [6.783]	-0.024** (0.009) [2.570]	-0.070** (0.034) [2.093]	-0.094*** (0.033) [2.866]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	Reg.	Reg.	Reg.	Reg.	Reg.
Baseline Y	25.755	18.244	0.015	0.162	0.176
Reduced form	-5.376***	-6.035***	-0.013***	-0.039**	-0.053***
First stage	0.570***	0.561***	0.561***	0.561***	0.561***
F-statistic	84.257	76.984	76.984	76.984	76.984
VtF 95% CV	1.918	1.939	1.912	1.915	1.912
#[Clusters]	445	445	445	445	445
#[Treated]	78,797	80,271	80,271	80,271	80,271
#[Untreated]	1,279,617	1,291,985	1,291,985	1,291,985	1,291,985
N	8,497,491	8,605,362	8,605,362	8,605,362	8,605,362
	Stats/Pats At Chosen Center (6)	Nurs./Pats At Chosen Center (7)	1[Hosp.] (8)	1[URR <65%] (9)	Expected Utility (MTEs) (10)
1[Has Center]	0.079*** (0.019) [4.096]	0.033*** (0.006) [5.967]	-0.018 (0.012) [1.442]	-0.006 (0.024) [0.246]	10.560*** (0.853) [12.383]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	Reg.	Reg.	Reg.	Reg.	Reg.
Baseline Y	0.227	0.060	0.182	0.101	54.122
Reduced form	0.045***	0.017***	-0.010	-0.003	6.028***
First stage	0.570***	0.502***	0.556***	0.450***	0.571***
F-statistic	84.257	32.962	72.405	27.944	84.692
VtF 95% CV	1.926	1.868	1.911	1.838	1.910
#[Clusters]	445	391	442	403	444
#[Treated]	78,797	31,496	66,131	36,536	78,797
#[Untreated]	1,279,622	719,041	1,043,137	682,411	1,276,451
N	8,497,561	3,285,988	6,936,889	3,615,376	8,453,167

Tab. A11. Sensitivity analysis of IV-DID estimates of marginal centers' effects on patient access, health, and welfare. This table reports a sensitivity analysis of the main results from the IV-DID analysis. Unlike the main results, these estimates are computed using regional comparison groups. The top panel reports estimates of β in equation (3a). County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). #[Treated] and #[Untreated] refer to unique patient-months and N refers to all observations in the stacked dataset. See the discussion near page 54. Source: author's analysis of the USRDS Database.

	Mi. to Chosen Center (1)	Mi. to Nearest Center (2)	1[HHD] (3)	1[PD] (4)	1[Home] (5)
1[Has Center]	-17.919*** (4.012) [4.466]	-21.750*** (5.572) [3.904]	-0.069* (0.041) [1.682]	-0.047 (0.060) [0.787]	-0.117* (0.067) [1.742]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	USA	USA	USA	USA	USA
Baseline Y	37.649	26.704	0.035	0.151	0.186
Reduced form	-6.728***	-7.734***	-0.025*	-0.017	-0.042*
First stage	0.375***	0.356***	0.356***	0.356***	0.356***
F-statistic	14.322	12.560	12.560	12.560	12.560
VtF 95% CV	2.416	2.291	1.716	1.725	1.715
#[Clusters]	1,371	1,371	1,372	1,372	1,372
#[Treated]	23,198	24,381	24,381	24,381	24,381
#[Untreated]	2,508,745	2,567,263	2,567,341	2,567,341	2,567,341
N	35,145,628	35,966,063	35,967,155	35,967,155	35,967,155
	Stats/Pats At Chosen Center (6)	Nurs./Pats At Chosen Center (7)	1[Hosp.] (8)	1[URR <65%] (9)	Expected Utility (MTEs) (10)
1[Has Center]	0.173*** (0.028) [6.082]	0.056*** (0.014) [4.139]	-0.071** (0.028) [2.527]	0.045* (0.024) [1.898]	17.982*** (3.222) [5.581]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	USA	USA	USA	USA	Reg.
Baseline Y	0.227	0.067	0.161	0.073	85.502
Reduced form	0.065***	0.020***	-0.025***	0.015**	5.966***
First stage	0.375***	0.363***	0.349***	0.335***	0.332***
F-statistic	14.322	14.206	11.533	10.737	10.296
VtF 95% CV	2.341	1.976	1.726	1.685	2.587
#[Clusters]	1,372	1,369	1,367	1,364	211
#[Treated]	23,198	18,729	20,426	17,141	23,198
#[Untreated]	2,508,828	1,967,762	2,065,431	1,707,865	187,955
N	35,146,790	27,567,397	28,936,460	23,927,251	2,654,568

Tab. A12. Sensitivity analysis of IV-DID estimates of marginal centers' effects on patient access, health, and welfare. This table reports a sensitivity analysis of the main results from the IV-DID analysis. Unlike the main results, these estimates are computed using only the 2007 WA reform. The top panel reports estimates of β in equation (3a). County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). Comparison groups are nationwide or regional counties without centers on the events' effective dates. #[Treated] and #[Untreated] refer to unique patient-months and N refers to all observations in the stacked dataset. See the discussion near page 54. Source: author's analysis of the USRDS Database.

	Mi. to Chosen Center (1)	Mi. to Nearest Center (2)	1[HHD] (3)	1[PD] (4)	1[Home] (5)
1[Has Center]	-6.396*** (1.064) [6.012]	-7.741*** (1.114) [6.950]	-0.012* (0.006) [1.819]	-0.055 (0.043) [1.280]	-0.067 (0.043) [1.563]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	USA	USA	USA	USA	USA
Baseline Y	20.734	14.497	0.005	0.167	0.172
Reduced form	-3.882***	-4.697***	-0.007*	-0.033	-0.040
First stage	0.607***	0.607***	0.607***	0.607***	0.607***
F-statistic	99.962	99.356	99.356	99.356	99.356
VtF 95% CV	1.937	1.953	1.924	1.924	1.924
#[Clusters]	1,728	1,728	1,729	1,729	1,729
#[Treated]	55,599	55,890	55,890	55,890	55,890
#[Untreated]	5,836,053	5,957,675	5,957,804	5,957,804	5,957,804
N	30,701,490	31,417,242	31,417,951	31,417,951	31,417,951
	Stats/Pats At Chosen Center (6)	Nurs./Pats At Chosen Center (7)	1[Hosp.] (8)	1[URR <65%] (9)	Expected Utility (MTEs) (10)
1[Has Center]	0.088*** (0.022) [4.066]	0.025*** (0.002) [15.792]	-0.007 (0.014) [0.484]	-0.034 (0.030) [1.123]	10.014*** (0.888) [11.273]
ZCTA FE	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y
Pat. Chars.	Y	Y	Y	Y	Y
Comp. group	USA	USA	USA	USA	Reg.
Baseline Y	0.226	0.051	0.192	0.127	53.294
Reduced form	0.054***	0.017***	-0.004	-0.018	6.652
First stage	0.607***	0.685***	0.601***	0.539***	0.664
F-statistic	99.963	166.009	87.535	22.924	106.407
VtF 95% CV	1.927	1.960	1.919	1.816	1.960
#[Clusters]	1,729	1,405	1,728	1,495	233
#[Treated]	55,599	12,767	45,705	19,395	55,599
#[Untreated]	5,836,219	2,843,894	4,826,845	2,863,982	1,088,496
N	30,702,437	5,781,992	25,750,152	10,216,501	5,798,599

Tab. A13. *Sensitivity analysis of IV-DID estimates of marginal centers' effects on patient access, health, and welfare.* This table reports a sensitivity analysis of the main results from the IV-DID analysis. Unlike the main results, these estimates are computed using only the NC threshold-crossings. The top panel reports estimates of β in equation (3a). County-level cluster-robust standard errors in parenthesis and $|t|$ in square brackets. Baseline Y is the average value of the outcome in the event counties prior to their events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). Comparison groups are nationwide or regional counties without centers on the events' effective dates. #[Treated] and #[Untreated] refer to unique patient-months and N refers to all observations in the stacked dataset. See the discussion near page 54. Source: author's analysis of the USRDS Database.

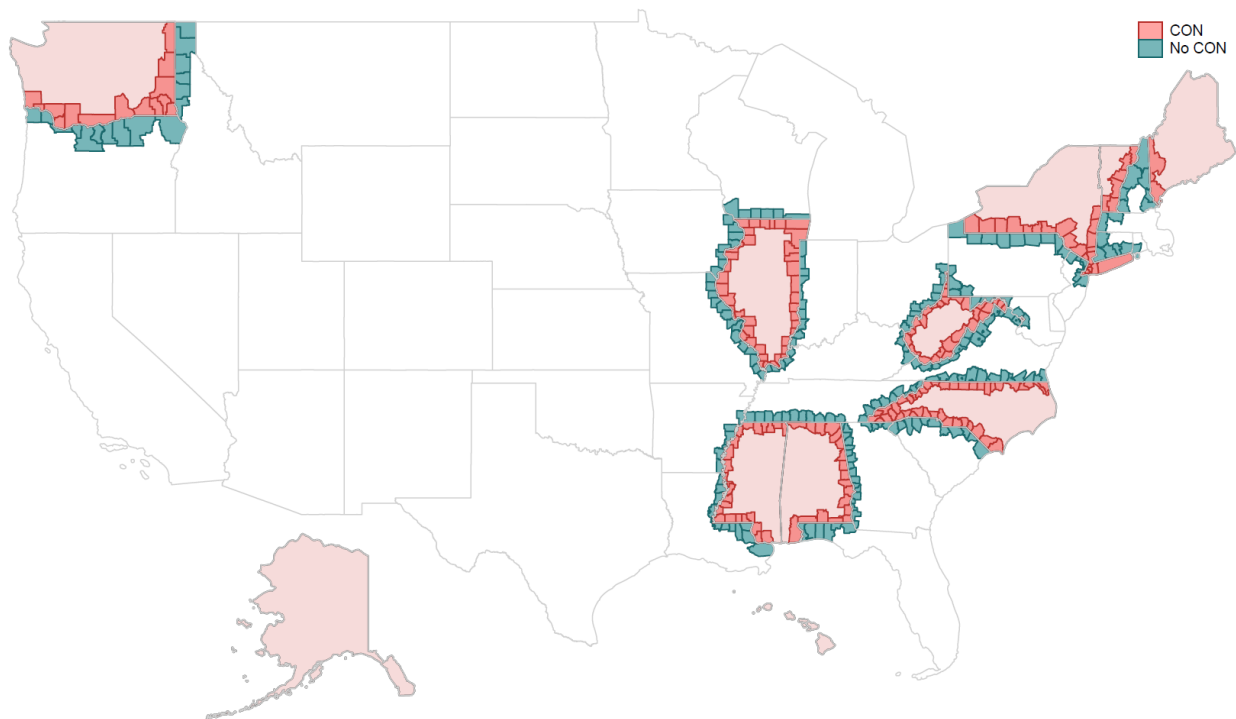


Fig. A1. *Cross-sectional variation in dialysis CON indicator.* This map identifies the states with and without dialysis CON programs during 2004-2016. It also identifies counties on the border between CON and non-CON states. See the discussion near page 9. Source: AHPA surveys in 2004, 2005, 2010, 2012, and 2016.

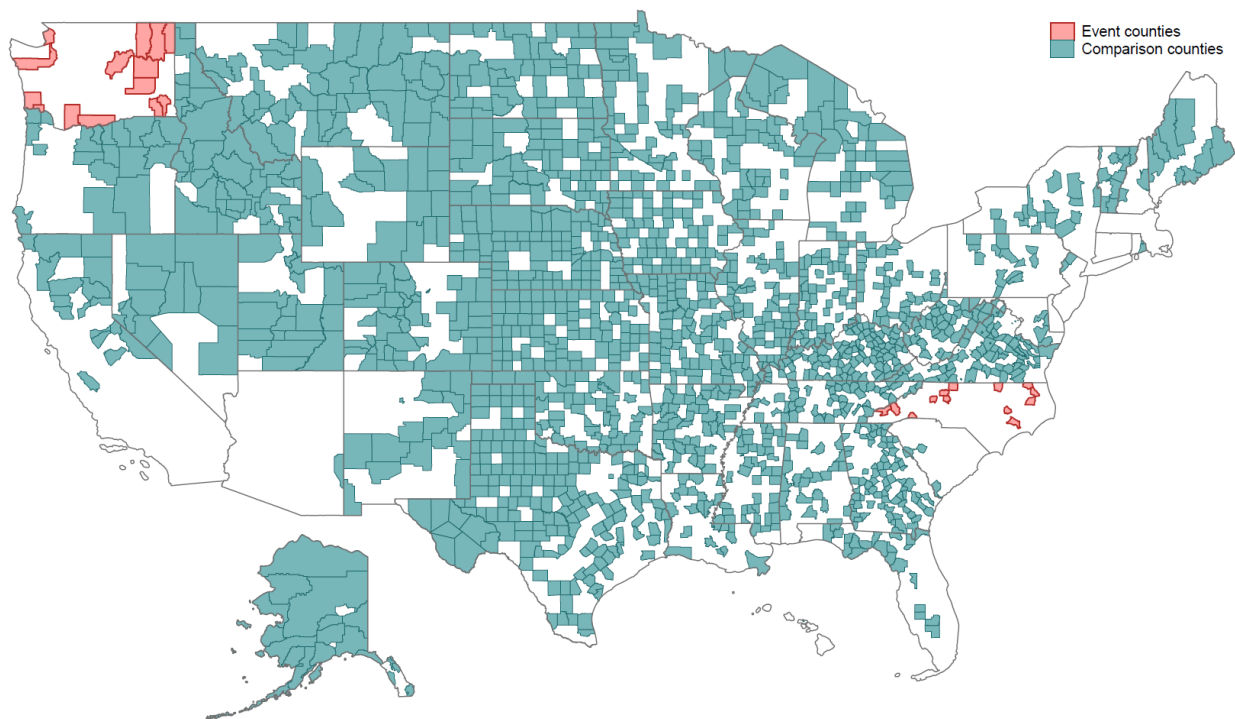
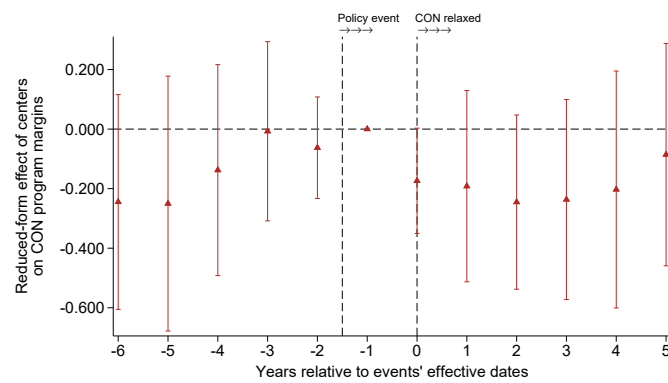
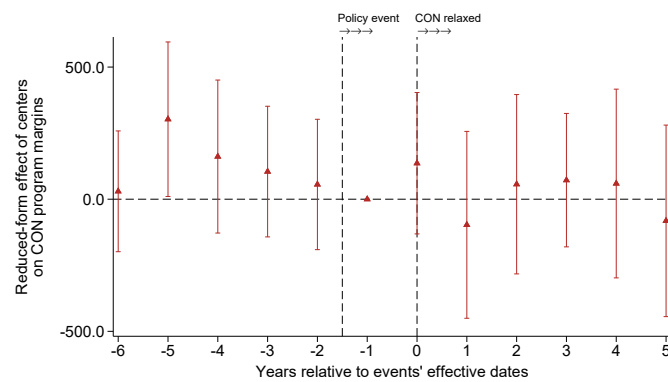


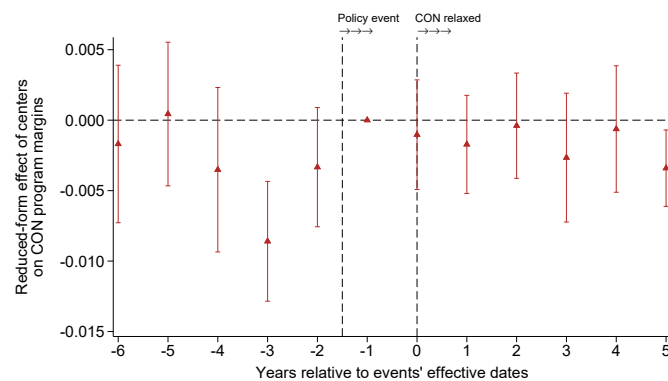
Fig. A2. WA and NC event counties. This map identifies the WA and NC counties where events statutorily relaxed CON-related entry barriers. It also identifies other counties in the U.S. that had zero centers at the time of one or more of the events, and that were consequently used in the comparison group of the IV-DID analysis. See the discussion near page 13. Source: author's analysis of the USRDS Database.



(a) HD sessions among HD patients



(b) Medicare spending among patients with Medicare FFS as a primary payer

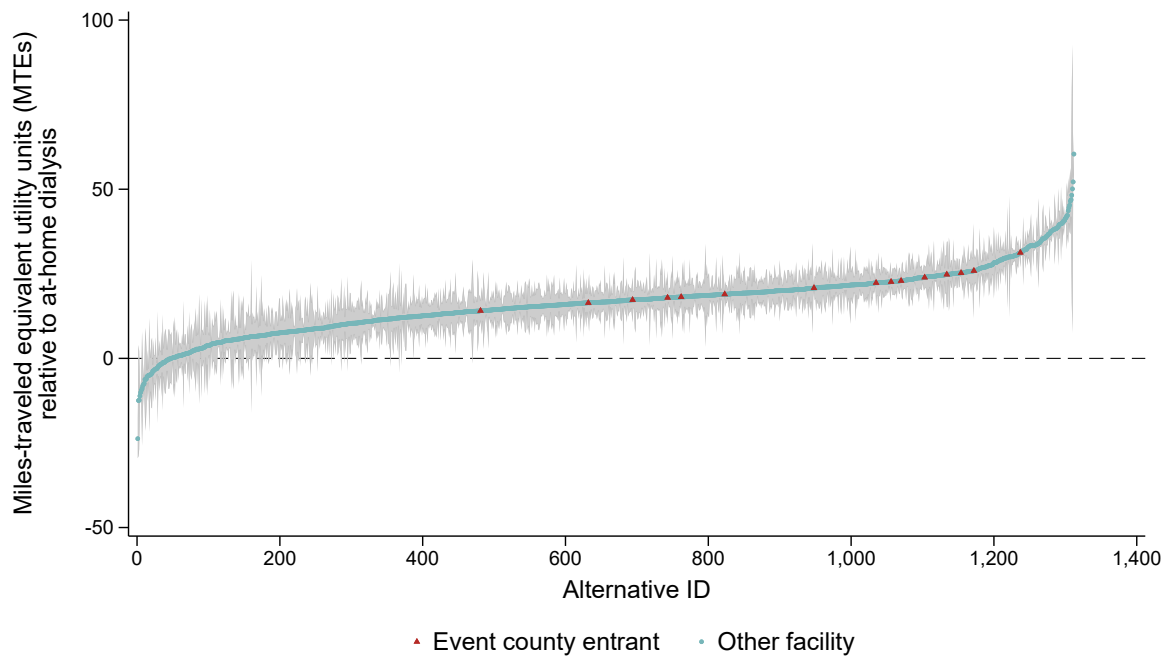


(c) 1[Died]

Fig. A3. *DID estimates of the effects of relaxing CON entry barriers on additional outcomes.* This figure reports additional reduced-form results of the IV-DID analysis. Each panel plots estimates of equation (4) for various outcomes. Bars are 95% confidence intervals estimated with county-level cluster-robust standard errors. All models include patient characteristics (age, race, sex, ethnicity, primary diagnosis, payer, and dialysis start year). Comparison groups are nationwide counties without centers on the events' effective dates. See the discussion near page 53. Source: author's analysis of the USRDS Database.

	(1)
TrDist	-0.129*** (0.005)
TrDist ²	0.000*** (0.000)
1[Stay]	5.878*** (0.036)
#[Cases]	1,368,224
#[Alts.]	1,312
<i>N</i>	27,881,747
Clusters	447
#[Alts. per case]	
Avg.	20
Min.	2
Max.	93

(a) Travel and switching cost estimates



(b) Distribution of alternative-specific fixed effect estimates

Fig. A4. *Sensitivity analysis of estimates of treatment choice model.* Panel (a) reports estimates of γ_1 , γ_2 , and λ in equation (G.5). County-level cluster-robust standard errors in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel (b) plots the distribution of alternative-specific fixed effects scaled by $-\hat{\gamma}_1$. Shaded areas are 95% confidence intervals. See the discussion near page 54. Source: author's analysis of the USRDS Database.